# HIGHER HIDA THEORY AND THE $p$-ADIC $L$-FUNCTION FOR $\mathrm{U}(2,1)$ (DRAFT)

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Abstract. We construct the $p$-adic $L$-function of certain *generic automorphic forms* of $\mathrm{GU}(2,1)$, following a general strategy suggested by [LPSZ]. To achieve this, we use the higher Hida theory for $\mathrm{U}(2,1)$ developed by M. T. Nguyen [Ng], complemented with several other ingredients necessary for our purpose. The integral formula of Gelbart–Piatetski-Shapiro is interpreted as a cohomological period integral, namely a certain $L$-function realized as a cup product in coherent cohomology of Shimura varieties. Towards a systematic generalization of this period integral, we develop the theory of *generalized Whittaker models*, into which many period integral can unfold. Examples include certain Rankin–Selberg integrals with non-unique models, and Gan–Gross–Prasad type period integrals for $\mathrm{U}(n,1)\times\mathrm{U}(n-1,1)$.

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## 1. Introduction

In view of Kato's explicit reciprocity law [Ka], there has been a constant need of a "de Rham" construction of $p$-adic $L$-functions. The study of $p$-adic variation of automorphic forms in the de Rham or coherent cohomology context has been extremely limited. In particular, most results can only deal with holomorphic automorphic forms, namely those whose infinity type belongs to holomorphic (limit of) discrete series.

On the other hand, a very recent development of Pilloni has come up with the notion of *higher Hida/Coleman theory*, which $p$-adically interpolates the higher coherent cohomology groups of Shimura varieties. These opened a way to approach for a de Rham construction of $p$-adic $L$-functions for automorphic representations that are not holomorphic. Indeed, such strategy was realized in [LPSZ] for the spin $L$-function of $\mathrm{GSp}_4$.

Our objective is to use the similar strategy to $p$-adically interpolate the base change $L$-function for generic forms of $\mathrm{U}(2,1)$. Our main theorem is as follows.

**Theorem 1.** *Suppose $\Pi$ is a generic, stable, cuspidal automorphic representation of $\mathrm{GU}(2,1)$ of infinitesimal character $(k_1,k_2,k_3;c)$ (see §2). If $k_1-k_2\geq 2$ and $k_2+k_3\geq 2$, there is an element $\mathcal{L}(\Pi,\mathbf{j})\in\Lambda_{\overline{\mathbb{Q}}_p}(\mathbb{Z}_p^\times)$, the Iwasawa algebra, such that, for any Dirichlet character $\chi$ of $p$-power conductor and any critical value $a$ of $L(\Pi,\chi,s)$,*

$$\mathcal{L}(\Pi,a+\chi)=Z_\infty(\Pi,\chi,a)Z_p(\Pi,\chi,a)L(BC(\Pi)\otimes\chi^{-1},a),$$

*where $Z_\infty$ and $Z_p$ are certain local factors at $\infty$ and $p$, respectively.*

As in [LPSZ], the two main ingredients are

- *cohomological period integrals* (period integrals relating values of $L$-functions to coherent cohomology of Shimura varieties), and
- the higher Hida theory.

In our case, the cohomological period integral in use is the one of Gelbart–Piatetski-Shapiro [GPS]. On the other hand, most of the higher Hida theory for $\mathrm{GU}(2,1)$ is developed in [Ng]. We expect the same strategy will work for the Gan–Gross–Prasad type period integral for $\mathrm{U}(n,1)\times\mathrm{U}(n-1,1)$ for cusp forms of infinity type (generic, holomorphic).

1.1. **Higher Hida theory for $\mathrm{U}(2,1)$.** To make use of the cohomological interpretation of a period integral, we need to interpolate the classes appearing in higher coherent cohomology, which is exactly what higher Hida theory does. Fortunately, the analogue of higher Hida theory

was developed in a Lyon thesis of M. T. Nguyen in [Ng]. The situation for $\mathrm{GU}(2,1)$ is actually easier than that for $\mathrm{GSp}_4$, as the Hecke correspondences $T_p$ and $U_p$ are represented by a miniscule diagonal matrix, so that there is no need to compose two different Hecke correspondences. In particular, there are strengthenings in several parts, most notably a proof of integral control theorem; this is in contrast with [Pi], where the integral control theorem is only proved only for a sufficiently regular weight (with inexplicit bounds on the regularity). As a result, [Ng] did not have to invoke automorphic representations in the course of proof of control theorem. We complement this with several calculations and results relating automorphic representations and the ordinary coherent cohomology.

1.2. **Generalized Whittaker models: Unfolding of period integrals.** The period integral we need for the purpose of this paper is already developed in [GPS]. On the other hand, as in the case of Novodvorsky's integral used in [LPSZ], this period integral is a certain degenerate case of a partly conjectural formula of Ichino–Ikeda type (also called the refined Gan–Gross–Prasad conjecture). Although the Ichino–Ikeda formula is now established by an amazing recent development by W. Zhang and his collaborators (cf. [Zh], [BPC+], [BPL+]), the proof of the formula is extremely roundabout. Furthermore, the Ichini–Ikeda formula is about the *norm* of the Gan–Gross–Prasad period integral, so the formula does not completely characterize the value of the period integral.

We show, on the other hand, that the period integral itself unfolds nicely into a certain local model, just as the classical Rankin–Selberg integral unfolds into Whittaker model, so that the period integral can be manifestly seen as an Euler product of local factors. The local model we use is the so-called *generalized Whittaker model*, in the sense of Japanese school of mathematicians (Shintani, Yamashita, Murase–Sugano, $\cdots$). We use this name to distinguish ourselves from using the *degenerate Whittaker models*, the spaces of functionals transforming under a smaller unipotent subgroup, which are typically infinite-dimensional as soon as one takes the unipotent group smaller than that appears on the wave-front set. Such infinite-dimensionality can be salvaged by also considering the action of a part of the Levi.

We also use freely the language of Fourier expansions with respect to closed subgroups. Typically the widely-known Fourier expansions of automorphic forms are Whittaker–Fourier expansion, $q$-expansion and Fourier–Jacobi expansion. The latter two only applies to holomorphic automorphic forms, whereas the former only applies to generic automorphic forms. The Whittaker–Fourier expansion is just the Fourier expansion associated to the maximal unipotent subgroup, $q$-expansion to the unipotent radical of the Siegel parabolic, and Fourier–Jacobi expansion to the unipotent radical of the "Jacobi parabolic." Making sense of more general Fourier expansions for more general types of automorphic forms requires overcoming some analytic issues coming from continuous spectrum.

Using the ingredients, we show that the generalized Whittaker model associated to the Jacobi parabolic of $\mathrm{U}(2,1)$ can be used to show that the GGP period integral for $\mathrm{U}(2,1)\times\mathrm{U}(1,1)$ is Eulerian; the local zeta integral is an integral of local generalized Whittaker function (of $\mathrm{U}(2,1)$-form) times a Whittaker function (of $\mathrm{U}(1,1)$-form). We also explain how these lines of ideas can explain the mystery of *Rankin–Selberg integrals with non-unique models*. This characterization of Eulerianity would be useful for the computation of local factors (especially at $p$ and $\infty$).

Finally, it would be extremely interesting to understand how the generalized Whittaker model interacts with a more general GGP period integrals. On one hand, the unfolding involves a

branching of unipotent group into a smaller unipotent group. On the other hand, the unfolding uses a Fourier expansion, which involves the wave-front of the involved representations. It is natural to expect that the group-theoretic nature of both ingredients would interact in an interesting way. For example, it would be interesting to discover an example of unfolding of a GGP period integral into generalized Whittaker models for a pair of non-generic representations.

1.3. **Prospects: The square-root $p$-adic $L$-function for $\mathrm{U}(n,1)\times\mathrm{U}(n-1,1)$.** We expect that the strategy developed in this paper can be ultimately applied to the construction of the "square-root $p$-adic $L$-function" for $\mathrm{U}(n,1)\times\mathrm{U}(n-1,1)$, for automorphic forms of infinity type (generic, holomorphic), as follows.

- Unfolding of the GGP period integral into generalized Whittaker model, in turn, gives rise to integral formula for the local factors of the integral, which is a more refined information that what the Ichino–Ikeda formula conveys. Thus, the local factors can be explicitly computed given explicit formulae for generalized Whittaker functions. The archimedean generalized Whittaker functions for $\mathrm{U}(n,1)$ are studied in [Ts1, Ts2, Ts3], whereas the nonarchimedean generalized Whittaker functions are studied in [MS1]. Indeed, we have computed the unramified local factors for $\mathrm{U}(2,1)\times\mathrm{U}(1,1)$, and the full computation of all local factors will be contained in a forthcoming work of the author.
- The GGP period integral for $\mathrm{U}(n,1)\times\mathrm{U}(n-1,1)$ would unfold similarly to the unfolding process of that for $\mathrm{U}(2,1)\times\mathrm{U}(1,1)$ dealt in §5. The main analytic difficulty does not arise, as the extra reductive part for the generalized Whittaker function is compact (namely, just $\mathrm{U}(1)$).
- The higher Hida theory for $\mathrm{GU}(n,1)$ would enjoy many simplifications. Firstly, as in the case of $\mathrm{GU}(2,1)$, the Hecke operators $T_p$ and $U_p$ would be represented as a single cohomological correspondence given by the parahoric level Shimura vareity. Moreover, even though the $\mathrm{GU}(n,1)$-Shimura variety would have many more Ekedahl–Oort strata, the stratification has a nested structure. Namely, there are $n+1$ strata, and the closure relation is a total order, so that there is unique closed stratum, which is contained in the closure of another stratum, which is contained in the closure of another stratum, etc., until you reach the unique open stratum, the ordinary locus. Thus, one can extend the argument used in the higher Hida theory for $\mathrm{GU}(2,1)$ to go through an induction over the depth of strata using generalized Hasse invariants (established in [Bo]). Similarly, the embedding of Igusa towers for the $\mathrm{GU}(n-1,1)$-Shimura variety to that of the $\mathrm{GU}(n,1)$-Shimura variety can be given completely analogously.

1.4. **Summary.** In §2, we set the basic notations. Then, in §3, we review the higher Hida theory for $\mathrm{U}(2,1)$ developed in [Ng]. In the course of review, we complement with several missing ingredients necessary for our development in this paper. In §4, we define the $p$-adic pushforward map from the smaller Shimura variety to the bigger Shimura variety, by specifying the embedding of partial Igusa towers.

In §5, we develop another upshot of the paper, namely the use of *generalized Whittaker models* in the study of period integrals. We first review the integral representation of Gelbart–Piatetski-Shapiro for the base-change $L$-function for $\mathrm{U}(2,1)$. This can be regarded as a degenerate case of the Gan–Gross–Prasad period integral for $\mathrm{U}(2,1)\times\mathrm{U}(1,1)$, whose explicit relation to values of $L$-functions is given by the refined Gan–Gross–Prasad conjectures. We develop a way to directly unfold the period integral involving the generalized Whittaker models for the GGP period integral

for $\mathrm{U}(2,1)\times\mathrm{U}(1,1)$. We also show that the same strategy can be used for other Rankin–Selberg type period integrals that were not known to be unfolded to certain nice local model.

In §6, we consider how coherent cohomology of the $\mathrm{GU}(2,1)$-Shimura variety is related to automorphic forms for $\mathrm{GU}(2,1)$. We describe the period integral as well as $p$-adic interpolation of it in terms of cup product in coherent cohomology. We also describe the theory of *nearly holomorphic and overconvergent modular forms* in terms of coherent cohomology of automorphic vector bundles.

In §7, we explicitly calculate the local factors of the Gelbart–Piatetski-Shapiro integral. Important are the calculations of the factors at $p$ and at $\infty$. In §8, we gather all the ingredients we developed to construct the desired $p$-adic $L$-function.

## 2. Setup

2.1. **Imaginary quadratic field.** Once and for all we choose an embedding $\iota_\infty:\overline{\mathbb{Q}}\to\mathbb{C}$ and $\iota_p:\overline{\mathbb{Q}}\to\mathbb{C}_p$, connected via $\iota:\mathbb{C}\xrightarrow{\sim}\mathbb{C}_p$. Let $K$ be an imaginary quadratic field and $p$ be a rational prime that splits in $K$. By choosing a preferred embedding $\sigma:K\to\overline{\mathbb{Q}}$, one gets a positivity in imaginary direction (via $\sigma_\infty=\iota_\infty\circ\sigma$) and a preferred $\pi\mid p$ (via $\sigma_p=\iota_p\circ\sigma$); thus we have a decomposition $p=\pi\overline{\pi}$.

We have $\mathcal{O}_K\otimes_{\mathbb{Z}}\mathcal{O}_K=\mathcal{O}_Ke^+\oplus\mathcal{O}_Ke^-$, where $e^+$ and $e^-$ are the idempotents corresponding to the identity $\mathrm{id}:K\to K$ and the complex conjugation $\bar{\cdot}:K\to K$, respectively.

2.2. **Groups.** Let $V=Y_K\oplus W\oplus X_K$ be a skew-Hermitian $K$-space of dimension 3 with the skew-Hermitian form $\theta_{2,1}$ defined by

$$\theta_{2,1}=\begin{pmatrix}&&-1\\&\theta&\\1&&\end{pmatrix}.$$

Here, the bases are given by $X_K=Kx, Y_K=Ky, W=Kw$, respectively, and $\theta\in K^\times$ is totally imaginary with $i\sigma_\infty(\theta)>0$, where $i$ is the usual $i$ (not $-i$) in $\mathbb{C}$. Without loss of generality, we assume that $\mathrm{Tr}_{K/\mathbb{Q}}(\theta)\in\mathbb{Z}$. Let $\langle,\rangle:V\times V\to\mathbb{Q}$ be the alternating pairing defined by $\langle v,v'\rangle=\mathrm{Tr}_{K/\mathbb{Q}}(\theta_{2,1}(v,v'))$.

Let $G=\mathrm{GU}(V)$ be the $\mathbb{Q}$-group of unitary similitudes of $(V,\theta_{2,1})$ (namely, $g$ such that $g\theta_{2,1}\overline{g}^T=\nu(g)\theta_{2,1}$), and $G'=\mathrm{U}(V)$ be the unitary group of $(V,\theta_{2,1})$. Let $\nu:G\to\mathbb{G}_{m,\mathbb{Q}}$ be the similitude morphism. Let also be $H\subset G$ ($H'\subset G'$, respectively) be the stabilizers of $W$. Then, $H,H'$ acts on $V_H=Y_K\otimes X_K$. For simplicity, we may call $G,G',H,H'$ as $\mathrm{GU}(2,1),\mathrm{U}(2,1),\mathrm{GU}(1,1)\times_{\mathrm{GL}_1}\mathrm{GU}(1),\mathrm{U}(1,1)\times\mathrm{U}(1)$, respectively. Moreover, we use the notation $[A]=A(\mathbb{Q})\backslash A(\mathbb{A})$ for a $\mathbb{Q}$-group $A$.

2.3. **Lattices.** Denote by $\mathfrak{d}_K$ the different of $K$, then take a lattice $M=Y\oplus L\oplus X^\vee:=\mathcal{O}_Ky\oplus\mathcal{O}_Kw\oplus\mathfrak{d}_K^{-1}x$. Then, $\langle,\rangle_{2,1}|_M$ is $\mathbb{Z}$-valued, and $M\otimes_{\mathbb{Z}}\mathbb{Z}_p=:M_p$ is self-dual with respect to $\langle,\rangle_{2,1}$. Consider $\mathcal{G}=\mathrm{GU}(M_p)$ and $\mathcal{G}'=\mathrm{U}(M_p)$. Let $M_\Sigma:=e^+M_p$ and $M_{\Sigma^c}:=e^-M_p$ be $\mathbb{Z}_p$-modules. For $v=\pi,\overline{\pi}$, we have identifications

$$G(K_v)\xrightarrow{\sim,g\mapsto(g|_{e^+V},\nu(g))}\mathrm{GL}_3(K_v)\times K_v^\times,$$

$$G'(K_v)\xrightarrow{\sim,g\mapsto g|_{e^+V}}\mathrm{GL}_3(K_v),$$

and the same identification gives

$$\mathcal{G}(\mathcal{O}_v) \xrightarrow{\sim, g \mapsto (g|_{M_\Sigma}, \nu(g))} \mathrm{GL}_3(\mathcal{O}_v) \times \mathcal{O}_v^\times,$$

$$\mathcal{G}'(\mathcal{O}_v) \xrightarrow{\sim, g \mapsto g|_{M_\Sigma}} \mathrm{GL}_3(\mathcal{O}_v).$$

Let $\mathcal{P}' \subset \mathcal{G}'$ be the parahoric subgroup stabilizing the plane through $y, w$, and let $\mathcal{P} = \mathbb{G}_m(\mathbb{Z}_p) \times \mathcal{P}'$. We will colloquially refer it as a *mirahoric subgroup*. We also consider the upper triangular Borels $B'_G \subset G'$, $B_G \subset G$, $B'_H \subset H'$ and $B_H \subset H$, defined as the stabilizer of $Y_K$.

2.4. **Shimura varieties.** Let $\Gamma = \Gamma^p \Gamma_p \subset G(\mathbb{A}_f)$ be a neat open compact subgroup. We will be only considering the two cases, $\Gamma_p = \mathcal{G}$ (hyperspecial) or $\Gamma_p = \mathcal{P}$ (mirabolic). Consider the moduli problem $Y_G(\Gamma) : \mathrm{Sch}_{\mathcal{O}_{K,\pi}} \to \mathrm{Sets}$ parametrizing the equivalence classes of quadruples $(A/S, i, \overline{\lambda}, \eta^p(, C))$ where

- $A/S$ is an abelian scheme of relative dimension 3 for a $\mathcal{O}_{K,\pi}$-scheme $S$,
- $i : \mathcal{O}_K \hookrightarrow \mathrm{End}_S(A) \otimes_{\mathbb{Z}} \mathbb{Z}_{(p)}$ satisfying the Kottwitz determinant condition,

$$\det(X - i(k) | \operatorname{Lie} A) = (X - \sigma(c(b)))^2 (X - \sigma(b)) \in \mathcal{O}_{K,\pi}[X], \quad \text{for all } k \in \mathcal{O}_K,$$

- $\lambda$ is a prime-to-$p$ polarization of $A$ over $A^\vee$ and $\overline{\lambda}$ is the $\mathbb{Z}_{(p),+}$-orbit of $\lambda$, inducing a Rosati involution, sending $i(k)$ to $i(\overline{k})^\vee$ for all $k \in \mathcal{O}_K$,
- $\eta^p$ is a $\Gamma^p$-level structure in the sense of [Ko],
- and if $\Gamma_p = \mathcal{P}$, then $C$ is an order $p^2$ subgroup of $A[\pi]$.

We tend to omit $\Gamma$ from the notation and rather write as $Y$ (for hyperspecial level at $p$) and $Y_{\mathrm{Kl}}$ (for mirahoric level at $p$) if there is no confusion.

There exists a universal abelian scheme $\mathcal{A}$ over $Y_G$, and the Hodge bundle $\omega = e^*_{\mathcal{A}/Y_G} \Omega^1_{\mathcal{A}/Y_G}$, where $e_{\mathcal{A}/Y_G}$ is the unit section, is a rank 3 bundle with the action of $\mathcal{O}_K$. Thus, it decomposes into $\omega = \omega_\sigma \oplus \omega_{\overline{\sigma}}$, according to the CM types; $\omega_\sigma$ is of rank 2, whereas $\omega_{\overline{\sigma}}$ is of rank 1, by the signature condition.

It is known that the moduli problem is represented by a quasiprojective $\mathcal{O}_{K,\pi}$-scheme, denoted also as $Y_G(\Gamma)$, which is also smooth when the level at $p$ is hyperspecial. Via $H \to G$, there is a natural embedding $\iota : Y_H(\Gamma_H) \to Y_G(\Gamma)$ for $\Gamma_H = \Gamma \cap H(\mathbb{A}_f)$ and $Y_H(\Gamma_H)$ is the scheme representing a similar moduli problem for $H$. This is on the other hand just a product of a modular curve and a definite unitary Shimura variety (thus a finite scheme); they are identified via taking the product of the two identical elliptic curves (from the modular curve) and a CM elliptic curve (from the definite unitary Shimura variety).

These Shimura varieties admit arithmetic toroidal compactifications, and the datum can be chosen so that the embedding extends to that of compactifications. Note that these are $\mathbb{Z}_p$-smooth if $\Gamma_p = \mathcal{G}$ but not so in general. We ambiguously denote a choice of nice toroidal compactification by $X_G(\Gamma)$, etc. We also denote the boundary divisor as $D$, if there is no confusion in notation. Note that we do not need to make any choices for $H$ as $Y_H$ is basically a copy of several modular curves. The choices will not matter because the cohomology does not depend on the choice. Also, we denote the minimal compactifications by $X_G(\Gamma)^*$, etc.

2.5. **Complex uniformization: two different ways.** We will talk about automorphic cohomology, so it is beneficial to write down the details of Cartan decomposition $\mathfrak{g} = \mathfrak{k} \oplus \mathfrak{p}$ and complex structure $\mathfrak{p}_{\mathbb{C}} = \mathfrak{p}_+ \oplus \mathfrak{p}_-$. Note first that as the center gets modded out in the construction of symmetric spaces, it is usually the simplest to work with special unitary groups. Various parts of the section came from [La], [He].

2.5.1. *Generalities.* Let us abstractly summarize what happens. We freely use standard notations that are not necessarily compatible with what are used in the other paragraphs. Given a semisimple algebraic group $G$ with $K \subset G(\mathbb{R})$ a maximal compact, as we have $\operatorname{rank}_{\mathbb{R}} K = \operatorname{rank}_{\mathbb{R}} G(\mathbb{R})$, there is a compact Cartan $T_c \subset K \subset G(\mathbb{R})$. From the choice of a Cartan involution, one has a Cartan decomposition $\mathfrak{g}_{\mathbb{R}} = \mathfrak{k}_{\mathbb{R}} \oplus \mathfrak{p}_{\mathbb{R}}$ (we have put the subscripts to distinguish them from their complexifications). Furthermore, the choice of a Hodge cocharacter, or equivalently the choice of the complex structure on $G(\mathbb{R})/K$, gives a decomposition $\mathfrak{p}_{\mathbb{C}} = \mathfrak{p}_+ \oplus \mathfrak{p}_-$. Let $P_-(\mathbb{C}) \subset G(\mathbb{C})$ be the parabolic subgroup whose Lie algebra is $\mathfrak{k}_{\mathbb{C}} \oplus \mathfrak{p}_-$. Then the obvious inclusion $G(\mathbb{R})/K \hookrightarrow G(\mathbb{C})/P_-(\mathbb{C})$ is an open immersion of complex manifolds.

Using the $P_-(\mathbb{C})$-torsor $G(\mathbb{C}) \to G(\mathbb{C})/P_-(\mathbb{C})$, for any algebraic representation $V$ of $P_-(\mathbb{C})$, one can form a holomorphic vector bundle $[V] = G(\mathbb{C}) \times^{P_-(\mathbb{C})} V \to G(\mathbb{C})/P_-(\mathbb{C})$, which restricts to a holomorphic vector bundle over $G(\mathbb{R})/K$ denoted also as $[V]$. It is known that these descend down to arithmetic quotients, and these admit arithmetic models. These are called *automorphic vector bundles.*

Here comes why we put the phrase "two different ways": when talking about automorphic vector bundles, or anything that comes out of using a compact Cartan, sometimes one works with a noncompact maximal torus, which is also the case for unitary groups. We claim that it does not really matter. Indeed, let $T \subset G$ be a maximal torus that is not necessarily $\mathbb{R}$-anisotropic. Still, $T(\mathbb{C})$ and $T_c(\mathbb{C})$ are conjugate, say $T(\mathbb{C}) = g^{-1}T_c(\mathbb{C})g$. Suppose further that there is a parabolic subgroup $T \subset P \subset G$ such that $P(\mathbb{C}) = g^{-1}P_-(\mathbb{C})g$. Then $G(\mathbb{C})/P_-(\mathbb{C}) \cong G(\mathbb{C})/P(\mathbb{C})$, $h \mapsto g^{-1}hg$. Using this, one can construct automorphic vector bundles for algebraic representations of $P(\mathbb{C})$. Furthermore, these constructions yield the same vector bundles as those using $P_-(\mathbb{C})$ and $K$, as algebraic representations of $P_-(\mathbb{C})$ and those of $P(\mathbb{C})$ are identified via the same conjugation. Also, later we will be interested in automorphic cohomology, in particular Lie algebra cohomology of real group representations, but then again the automorphic vector bundles built for $T$ are literally the same as automorphic vector bundles built for $T_c$, so the calculation of Lie algebra cohomology can be done purely in terms of $T_c \subset K \subset P_-$.

2.5.2. *In the case of* $\mathrm{U}(2,1)$. There are two prominent ways of dealing with unitary groups. One can work with the antidiagonal matrix as we do, but one can also work with the diagonal matrix. In our case, it would be to use the Hermitian diagonal matrix $1_{2,1} := \begin{pmatrix} 1 & & \\ & 1 & \\ & & -1 \end{pmatrix}$. Let $\mathrm{U}_{\mathrm{diag}}$ be the unitary group defined using $1_{2,1}$ (and similarly for other related groups). Then, $\mathrm{U}_{\mathrm{diag}}(\mathbb{R})$ is isomorphic to $\mathrm{U}(2,1)(\mathbb{R})$ via

$$\mathrm{U}_{\mathrm{diag}}(\mathbb{R}) \xrightarrow{g \mapsto cgc^{-1}} \mathrm{U}(2,1)(\mathbb{R}), \quad c = \begin{pmatrix} \frac{\sqrt{-k}}{2}(1+i) & & \frac{\sqrt{-k}}{2}(1+i) \\ & 1 & \\ \frac{\sqrt{-k}}{2}(-1+i) & & \frac{\sqrt{-k}}{2}(1-i) \end{pmatrix},$$

where $\theta = ik$. Namely,

$$c(\theta 1_{2,1})\overline{c}^T = \theta_{2,1}.$$

The unitary group in the form of $\mathrm{U}_{\mathrm{diag}}$ acts naturally on the complex ball $\mathcal{D} = \{(z_1, z_2) \in \mathbb{C}^2 \mid |z_1|^2 + |z_2|^2 < 1\}$ via $g \in \mathrm{U}_{\mathrm{diag}}$ acting on $1 \times 3$ matrix $\begin{pmatrix} z_1 & z_2 & 1 \end{pmatrix}$, namely

$$\begin{pmatrix} z & 1 \end{pmatrix} \begin{pmatrix} A & B \\ C & D \end{pmatrix} = \begin{pmatrix} zA + C & zB + D \end{pmatrix} \sim \begin{pmatrix} \frac{zA+C}{zB+D} & 1 \end{pmatrix}, \quad z = \begin{pmatrix} z_1 & z_2 \end{pmatrix}.$$

A natural choice of the maximal compact subgroup is the stabilizer of $0 \in \mathcal{D}$, namely $K_{\infty,\mathrm{diag}} := \begin{pmatrix} \mathrm{U}_2 & \\ & \mathrm{U}_1 \end{pmatrix}$. From this, we see that $K_{\infty,\mathbb{C}}$ is conjugate to $M_P(\mathbb{C})$.

The unitary group in the form of $\mathrm{U}(2,1)$ acts naturally on the unbounded domain

$$X = \left\{ \tau = \begin{pmatrix} x & y \end{pmatrix} \in M_{2\times 1}(\mathbb{C}) \mid 2\sqrt{-k}\,\mathrm{Im}\, x > |y|^2 \right\},$$

via $g \in \mathrm{U}(2,1)$ acting on $3 \times 1$ matrix $\begin{pmatrix} \tau \\ 1 \end{pmatrix}$, namely

$$\begin{pmatrix} A & B \\ C & D \end{pmatrix} \begin{pmatrix} \tau \\ 1 \end{pmatrix} = \begin{pmatrix} A\tau + B \\ C\tau + D \end{pmatrix}.$$

A natural choice of the maximal compact subgroup is the stabilizer of $\begin{pmatrix} i \\ 0 \end{pmatrix} \in X$, which corresponds to $K_{\infty,\mathrm{diag}}$ after the conjugation by $c$ as denoted above.

The Hodge cocharacters correponding to the complex structures of the above realizations of symmetric spaces are

$$h_{\mathrm{diag}} : S^1 \to \mathrm{U}_{\mathrm{diag}}, \quad h_{\mathrm{diag}}(z) = \begin{pmatrix} z & & \\ & z & \\ & & \overline{z} \end{pmatrix},$$

$$h : S^1 \to \mathrm{U}(2,1), \quad h(x+yi) = \begin{pmatrix} x & & y \\ & x+yi & \\ -y & & x \end{pmatrix}.$$

The Lie algebra of $\mathrm{U}_{\mathrm{diag}}$ is given by

$$\mathfrak{g}_{\mathrm{diag}} = \left\{ \begin{pmatrix} a & b & c \\ -\overline{b} & d & e \\ -\overline{c} & \overline{e} & f \end{pmatrix} \mid a, d, f \in i\mathbb{R} \right\},$$

$$\mathfrak{p}_{\mathrm{diag}} = \left\{ \begin{pmatrix} 0 & 0 & c \\ 0 & 0 & e \\ \overline{c} & \overline{e} & 0 \end{pmatrix} \right\}, \quad \mathfrak{k}_{\mathrm{diag}} = \left\{ \begin{pmatrix} a & b & 0 \\ -\overline{b} & d & 0 \\ 0 & 0 & f \end{pmatrix} \mid a, d, f \in i\mathbb{R} \right\}.$$

Under this, $\mathfrak{p}_{\mathrm{diag},\mathbb{C}} = \mathfrak{p}^+_{\mathrm{diag}} \oplus \mathfrak{p}^-_{\mathrm{diag}}$, where $\mathfrak{p}^-_{\mathrm{diag}}$ ($\mathfrak{p}^+_{\mathrm{diag}}$, respectively) is generated by $X^- = N^+ \otimes i - N^- \otimes 1$ and $Y^- = M^+ \otimes i - M^- \otimes i$ ($X^+ = N^+ \otimes i + N^- \otimes 1$, $Y^+ = M^+ \otimes i + M^- \otimes 1$, respectively), where

$$N^- = \begin{pmatrix} & i \\ -i & \end{pmatrix}, N^+ = \begin{pmatrix} & 1 \\ 1 & \end{pmatrix}, M^- = \begin{pmatrix} & i \\ -i & \end{pmatrix}, M^+ = \begin{pmatrix} & 1 \\ 1 & \end{pmatrix}.$$

2.6. **Automorphic vector bundles.** Let $T \subset G$, $T' \subset G'$ be the maximal torus consisted of diagonal matrices. Note that $T, T'$ can be identified with $K^\times \times K^\times$ and $K^\times \times K^1$, respectively, where $K^1 = \{z \in K^\times \mid z\overline{z} = 1\}$, via

$$T' = \left\{ \begin{pmatrix} a & & \\ & b & \\ & & \overline{a}^{-1} \end{pmatrix} : a \in K^\times, b \in K^1 \right\},$$

and similarly for $T$. Let $Z$ be the center of $\mathrm{GU}(2,1)$, identified with $\mathbb{G}_m(\mathbb{Q})$. Then these fit into an exact sequence

$$1 \to \mu_2 \to Z \times T' \to T \to 1.$$

The character group $X^*(T)$ is identified with $\{(k_1, k_2, k_3; c) \in \mathbb{Z}^4 \mid k_1 + k_2 + k_3 \equiv c (\mathrm{mod}\, 2)\}$ via

$$\lambda(k_1, k_2, k_3; c) : \begin{pmatrix} za & & \\ & zb & \\ & & z\overline{a}^{-1} \end{pmatrix} \mapsto \sigma(a)^{k_1}\sigma(b)^{k_2}\sigma(\overline{a})^{k_3} z^c.$$

It is $M_P$-dominant if $k_1 + k_3 \geq 0$ and is $G$-dominant if $k_1 \geq k_2$ and $k_2 + k_3 \geq 0$. We write $W_G(k_1, k_2, k_3; c)$ and $V_G(k_1, k_2, k_3; c)$ for representations of $M_P$ and $G$ with the corresponding highest weights, respectively. Note that $T$ is also a maximal torus of $H$, and $M_{B_H} = T$; we write $W_H(k_1, k_2, k_3; c)$ for the corresponding character of $T = M_{B_H}$. It is $H$-dominant if $k_1 \geq k_2$; we write $V_H(k_1, k_2, k_3; c)$ for the corresopnding $H$-representation.

The $\sigma$-part of the $\mathcal{O}_K$-action on the first de Rham cohomology sheaf $\mathscr{H}^1_{\mathrm{dR}}(\mathcal{A}/Y_G)$ has a Hodge filtration

$$0 \to \omega_\sigma \to \mathscr{H}^1_{\mathrm{dR}}(\mathcal{A}/Y_G)_\sigma \to \omega_{\overline{\sigma}}^\vee \to 0,$$

and the torsor of filtration-preserving trivializations of $\mathscr{H}^1_{\mathrm{dR}}(\mathcal{A}/Y_G)_\sigma$ gives a $P_{\mathcal{O}_{K,\pi}}$-torsor

$$\mathcal{T}_{G,\mathcal{O}_{K,\pi}}/Y_{G,\mathcal{O}_{K,\pi}}.$$

A similar construction works over an infinite place; $(\mathrm{Res}_{K/\mathbb{Q}}\, \mathcal{T})_\mathbb{R} \to (\mathrm{Res}_{K/\mathbb{Q}}\, Y)_\mathbb{R}$ can be thought as the restriction of the $P(\mathbb{C})$-torsor $G(\mathbb{C}) \to G(\mathbb{C})/P(\mathbb{C})$. Here, note that $M_P(\mathbb{R})$ is not a maximal compact (modulo center) subgroup of $G(\mathbb{R})$, so strictly speaking one cannot directly use $G(\mathbb{C})/P(\mathbb{C})$; on the other hand, as we will see, $M_P(\mathbb{C})$ is a $G(\mathbb{C})$-conjugate of $K_{\infty,\mathbb{C}}$, so after all we can work as if the diagonal matrix we have is a compact Cartan (modulo center).

Given an algebraic representation $V$ of $P$ over $\mathbb{Z}_p$, one defines an automorphic vector bundle $[V] := V \times^{P_{\mathcal{O}_{K,\pi}}} \mathcal{T}_{\mathcal{O}_{K,\pi}}$ over $Y_{G,\mathcal{O}_{K,\pi}}$. In particular, an $M_P$-highest weight representation $W_G$ can be made into an automorphic vector bundle $[W_G]$ which we also write as $\omega_G$, and similarly for $H$ as well. By considering what is center and what is not, we see that the underlying sheaf of $\omega_G$ ($\omega_H$, respectively) depends only on $(k_1 - k_2, k_2 + k_3)$ ($k_1 + k_3$, respectively). Thus we may want to only parametrize these parameters, and write $\omega_G(m_1, m_2) = \omega_G(-m_2, -m_1 - m_2, m_1 + 2m_2; -6)$ and $\omega_H(m) = \omega_H(0, -m, m; -2)$. The whole construction works equally well over toroidal compactification by using their canonical extensions (cf. [MP]).

Note that

$$\omega_G(k_1, k_2, k_3; k_1 + k_2 + k_3 - 6) \cong \mathrm{Sym}^{k_1 - k_2}\, \omega_\sigma \otimes (\det \omega_\sigma)^{k_2} \otimes \omega_{\overline{\sigma}}^{k_3}.$$

This reflects the fact that $K'_\infty = \mathrm{U}(2) \times \mathrm{U}(1)$. By our earlier observation, we see that $\det \omega_\sigma \cong \omega_{\overline{\sigma}}$. By Kodaira–Spencer, $\Omega^1_{X_G}(\log D_G)$ is the automorphic vector bundle corresponding to the adjoint representation on $\mathrm{Lie}\, N_P$. Thus $\Omega^1_{X_G}(\log D_G) \cong \omega_G(1,1)$, $\Omega^2_{X_G}(\log D_G) \cong \omega_G(0,3)$, $\Omega^1_{X_H}(\log D_H) \cong \omega_H(1,1)$. The conormal bundle to $Y_H \to Y_G$ corresponds to the automorphic

vector bundle associated with the character $\alpha_{G/H} = \omega_H(-1,2)$ of $M_{B_H}$. As in [LPSZ], there are pullback/pushforward maps of coherent cohomology of automorphic vector bundles of $H$ and $G$,

$$H^i(X_G, [V]) \xrightarrow{\iota^*} H^i(X_H, [V|_B]),$$

$$H^i(X_H, [V|_{B_H} \otimes \alpha_{G/H}^{-1}]) \xrightarrow{\iota_*} H^{i+1}(X_G, [V]),$$

and the corresponding maps for subcanonical extensions.

## 3. Higher Hida theory for $\mathrm{GU}(2,1)$

As a preliminary, we need to develop the analogue of higher Hida theory, as in [Pi] and [LPSZ, §3], for $\mathrm{GU}(2,1)$. A Lyon thesis of M. T. Nguyen [Ng] has generalized the higher Hida theory to a certain extent, which we will review and complement appropriately.

3.1. **Overview of** [Ng]**.** In their thesis [Ng], Nguyen directly generalized the works of [Pi] to $\mathrm{U}(2,1)$. For this to be useful for our purpose, we would need a control theorem for the range of weights where coherent $H^1$ is supposed to be the single degree of contribution (i.e. infinity type being generic discrete series). In the language of previous section, it is the range of $m_2 < 0$ and $m_1 + m_2 > 0$, or in the language of [Ng], it is the range of $k_2 + k_3 < 0$ and $k_1 + k_3 > 0$.

One would expect that the higher Hida theory for unitary groups with $p$ split is easier to develop, as the spherical Hecke algebra is generated by Hecke operators which are miniscule (namely, the double cosets generated by the elements of the form $\mathrm{diag}(p, p, \cdots, p, 1, \cdots, 1)$). Thus, the Hecke correspondences $T_p$ and $U_p$ are defined using only one geometric correspondence, and the analysis of $p$-divisibility is easier. This is indeed the case in [Ng]. Due to a simple nature of the correspondence, [Ng] proves *integral control theorems* for both $H^0$ and $H^1$ without much trouble. This contrasts with the situation of [Pi] (Theorem 1.1 of *op. cit.*), where the integral control theorem is only proven for weights satisfying a regularity hypothesis involving some unknown constant $C$, coming from Proposition 7.4.2.2 of *op. cit.*. To prove a characteristic zero control theorem for higher Hida theory, Pilloni uses the higher Coleman theory ([Pi, Theorem 14.8.1]). In particular, the classicality of ordinary classes in higher Coleman theory requires explicit calculations of Hecke eigenvalues of local representations; in *loc. cit.*, Proposition 5.1.5.2 is used in the proof of Lemma 14.8.3. Such discussion is unneccessary in [Ng]; on the other hand, such ingredient would be neccessary for our purpose, and we complement *op. cit.* with necessary ingredients. We closely follow the structure of [Pi].

3.1.1. *The $T$-operator.*

**Definition 3.1.** The naive Hecke operator $T_p^{naive}$ is defined as a cohomological correspondence ([Pi, §3]) $p_2^*(-) \to p_1^*(-) \to p_1^!(-)$ of

$$\begin{tikzcd} & X_{\mathrm{Kl}} \arrow[dl, "p_2"'] \arrow[dr, "p_1"] & \\ X & & X \end{tikzcd}$$

where $p_1(A/S, i, \overline{\lambda}, \eta^p, C) = (A/S, i, \overline{\lambda}, \eta^p)$, and $p_2(A/S, i, \overline{\lambda}, \eta^p, C)$ is induced from $A' = A/(C + C^\perp)$ and the polarization $\widetilde{\lambda} : A' \to (A')^t$ descended from $p\lambda : A \to A^t$.

This corresponds to the double coset operator

$$\mathcal{G}(\mathcal{O}_v)\,\mathrm{diag}(p,p,1)\mathcal{G}(\mathcal{O}_v).$$

Note that the cohomological correspondence of [Pi, §3] behaves well if the involved schemes are all Cohen–Macaulay, which is indeed the case by computing the local model of $X_{\mathrm{Kl}}$ ([Ng, Proposition 10.2.1]). In particular, $p_1, p_2$ are finite flat (Lemma 10.2.1 of *op. cit.*).

**Proposition 3.2** (Proposition 10.4.1 of *op. cit.*)**.** *If $k_1 + k_3 \geq 1$, $T_p^{naive}$ factorizes through*

$$p^{k_2+1} p_1^! \omega_G(k_1, k_2, k_3; k_1 + k_2 - k_3 - 4).$$

Here, we follow the rule set in §2.6 for the weight of central $\mathbb{G}_m$, so we will omit the $\mathbb{G}_m$-weight from now on. Accordingly, we define $T_p = \frac{1}{p^{k_2+1}} T_p^{naive}$.

3.1.2. *Finiteness of the ordinary cohomology.* Let $X_n$ be the $\mathbb{Z}/p^n\mathbb{Z}$-reduction of $X$, and let $X_1^{\leq i}$ be the locus where the multiplicative rank of $\mathcal{A}[\pi]$, the universal $p$-torsion group, is $\leq i$. This gives rise to the Ekedahl–Oort stratification of $X_1$,

$$X_1^{\leq 0} \subset X_1^{\leq 1} \subset X_1^{\leq 2} = X_1.$$

These admit a generalized Hasse invariant in an optimal way. Namely, the generalized Hasse invariant as a section of weight $(p^2-1, p^2-1, 0)$ on $X_1^{=1}$ extends to a section of the same weight on $X_1^{\leq 1}$ and vanishes to order 1 over $X_1^{=0}$. The $T_p$-operator interacts with multiplication by (generalized) Hasse invariants in an expected way, and using that one can show the following

**Proposition 3.3** (Proposition 10.6.3 and 11.2.3 of *op. cit.*)**.** *Let $e_{\mathrm{Kl}}$ be the ordinary projector for complexes associated with $T_p$. For $k_1 - k_2 \geq 1, k_1 + k_3 > 3$, $T_p$ acts trivially on $R\Gamma(X_1^{=0}, \omega(k_1 + (p^2+p-2), k_2 + (p^2+p-2), k_3))$. Moreover,*

$$e_{\mathrm{Kl}} R\Gamma(X_1, \omega_G(k_1, k_2, k_3)) \xrightarrow{\sim} e_{\mathrm{Kl}} R\Gamma(X_1^{\geq 1}, \omega_G(k_1, k_2, k_3)).$$

*Proof.* We say a word about the first statement, which is an analogue of [Pi, Proposition 7.4.2.2]. In [Pi], the analogous statement is proved for $k \geq C$ for an unknown yet universal constant $C > 0$. There, the $T_p$-operator involves three different levels, hyperspecial, Klingen and paramodular. In [Ng], only hyperspecial and Klingen levels appear, and the we know much more about the correspondence.

- $T_p$ is zero on $X_{\mathrm{Kl},1}^{=1,oo}$, corresponding to $H^\perp$ being biconnected. Thus, it is sufficient to prove that the restriction of $T_p$ to $X_{\mathrm{Kl},1}^{\leq 1,\text{ét}}$ vanishes on $X_{\mathrm{Kl},1}^{\leq 1,\text{ét}} \cap V(p_1^*\mathcal{I})$, where $X_{\mathrm{Kl},1}^{\leq 1,\text{ét}}$ is the closure of $X_{\mathrm{Kl},1}^{=1,\text{ét}}$ (corresponding to $H^\perp$ being étale) in $X_{\mathrm{Kl},1}^{\leq 1}$, and $\mathcal{I}$ is the ideal sheaf generated by $\mathrm{Ha}'$, defining $X_1^{=0}$ in $X_1^{\leq 1}$.
- The natural maps $p_1, p_2 : X_{\mathrm{Kl},1}^{\leq 1,\text{ét}} \to X_1^{\leq 1}$ are extremely easy. Firstly, $p_2$ is an isomorphism, as there is unique choice of étale subgroup $H^\perp \subset \mathcal{A}[\overline{\pi}]$ of order $p$. In this viewpoint, $p_1$ is $\mathrm{Frob}^2$. Thus, $V(p_1^*(\mathcal{I})) = p^2 V(p_2^*(\mathcal{I})) = p^2 V(p_2^* \mathrm{Ha}')$.
- The compatibility with generalized Hasse invariant is a commutative diagram

$$\begin{tikzcd}
p_2^*\omega(k_1+(p-1), k_2+(p-1), k_3) \arrow[r, "T_p"] \arrow[d, "\times p_2^*\mathrm{Ha}'"'] & p_1^!\omega(k_1+(p-1), k_2+(p-1), k_3) \arrow[d, "\times p_1^*\mathrm{Ha}'"] \\
p_2^*\omega(k_1+(p^2+p-2), k_2+(p^2+p-2), k_3) \arrow[r, "T_p"] & p_1^!\omega(k_1+(p^2+p-2), k_2+(p^2+p-2), k_3)
\end{tikzcd}$$

over $X^{\leq 1}_{\mathrm{Kl},1}$. From a simple nature of $p_1, p_2 : X^{\leq 1,\text{ét}}_{\mathrm{Kl},1} \to X^{\leq 1}_1$, over $X^{\leq 1,\text{ét}}_{\mathrm{Kl},1}$, we know that $p_1^* \operatorname{Ha}'$ and $p_2^* \operatorname{Ha}'$ vanishes along $V(p_2^*(\mathcal{I}))$ to order $p^2$ and 1, respectively. Thus,

$$T_p : p_2^*\omega(k_1 + (p^2+p-2), k_2 + (p^2+p-2), k_3) \to p_1^!\omega(k_1 + (p^2+p-2), k_2 + (p^2+p-2), k_3),$$

vanishes along $V(p_2^*(\mathcal{I}))$ to order at least $p^2 - 1$. One can compensate for an extra order of vanishing by using this fact for $(k_1 - 1, k_2, k_3)$ and realizing that the universal isogeny for $T_p$ for $(k_1, k_2, k_3)$ is that for $(k_1 - 1, k_2, k_3)$ tensored with the universal isogeny $\omega_{\mathcal{A}'[\pi^\infty]} \to \omega_{\mathcal{A}[\pi^\infty]}$, which vanishes at $V(p_2^*\mathcal{I})$ as well.

□

3.1.3. *Igusa tower and families of sheaves.* We now introduce the family of sheaves which will interpolate the classical ordinary cohomology.

**Definition 3.4** ([Ng, §12])**.** Let $\mathfrak{X}^{\geq 1}$ be the completion of $X$ along $X_1$. Let $\mathfrak{X}^{\geq 1}(p^m)$ be the formal moduli parametrizing $H_m \subset \mathcal{A}[\pi^\infty]$ étale-locally isomorphic to $\mu_{p^m}$. Over it, we define a partial Igusa tower $\mathfrak{IG}^{\geq 1}(p^m) = \operatorname{Isom}_{\mathfrak{X}^{\geq 1}(p^m)}(\mu_{p^m}, H_m)$. Let $\mathfrak{X}^{\geq 1}(p^\infty)$ be the inverse limit of $\mathfrak{X}^{\geq 1}(p^m)$, and let $\mathfrak{IG}^{\geq 1}(p^\infty) = \operatorname{Isom}_{\mathfrak{X}^{\geq 1}(p^\infty)}(\mu_{p^\infty}, H_\infty)$. Let $\pi : \mathfrak{IG}^{\geq 1}(p^\infty) \to \mathfrak{X}^{\geq 1}(p)$ be the projection. Let $\Lambda = \mathbb{Z}_p[[\mathbb{Z}_p^\times]]$ be the Iwasawa algebra and $\kappa : \mathbb{Z}_p^\times \to \Lambda^\times$ be the universal character. We define the sheaf

$$\mathfrak{F}^\kappa = (\pi_* \mathcal{O}_{\mathfrak{IG}^{\geq 1}(p^\infty)} \widehat{\otimes}_{\mathbb{Z}_p} \Lambda)[\Gamma = \kappa - k_2] \otimes \omega(k_2, k_2, k_3),$$

where $\Gamma = \mathbb{Z}_p^\times$. Given any character $\lambda : \mathbb{Z}_p^\times \to R^\times$, we define its specialization $\mathfrak{F}^\lambda := \mathfrak{F}^\kappa \widehat{\otimes}_{\Lambda,\lambda} R$.

One can also define the truncated Igusa tower (using $\mathfrak{IG}^{\geq 1}(p^m)$ instead of $\mathfrak{IG}^{\geq 1}(p^\infty)$) in a similar way.

3.1.4. *The $U$-operator and its relation with the $T$-operator.* Over $\mathfrak{X}^{\geq 1}_n(p^m)$, we can, in a somewhat ad hoc way, define the geometric correspondence that gives rise to the $U_p$-operator.

**Definition 3.5** ([Ng, §12.3])**.** Let $\mathfrak{C}_{\mathfrak{Y}}(p^m)$ be the formal moduli over $\mathfrak{Y}^{\geq 1}(p^m)$ parametrizing $L \subset \mathcal{A}[\pi]$ of order $p^2$ such that $L \cap H_m = 0$. This lies in a correspondence diagram

$$\begin{tikzcd} & \mathfrak{C}_{\mathfrak{Y}}(p^m) \arrow[dl, "p_2"'] \arrow[dr, "p_1"] & \\ \mathfrak{Y}^{\geq 1}(p^m) & & \mathfrak{Y}^{\geq 1}(p^m) \end{tikzcd}$$

where $p_1$ forgets $L$ and $p_2$ sends $(L, H_m, A)$ to $(\frac{H_m + L + L^\perp}{L + L^\perp}, \frac{A}{L + L^\perp})$. These can be compactified to

$$\begin{tikzcd} & \mathfrak{C}(p^m) \arrow[dl, "p_2"'] \arrow[dr, "p_1"] & \\ \mathfrak{X}^{\geq 1}(p^m) & & \mathfrak{X}^{\geq 1}(p^m) \end{tikzcd}$$

**Lemma 3.6** ([Ng, Lemma 12.3.1])**.** *The formal scheme $\mathfrak{C}(p^m)$ is Cohen–Macaulay, and $p_1$ is finite flat.*

Using this, we can define the $U_p$-operator.

**Definition 3.7.** Let $U_p^{naive}$ be the cohomological correspondence defined by the above correspondence diagram, and $U_p = \frac{1}{p^{k_2+1}} U_p^{naive}$.

As in Pilloni's work, the spherical ordinary cohomology and the mirahoric ordinary cohomology are geometrically related over the special fiber.

**Theorem 3.8** ([Ng, §12-§13])**.**

*(1) Over the mod $p$ ordinary locus $X_1^{=2}$, $T_p = T_p^{1,1}$, which only accounts for the multiplicative rank $1$ locus of $X_1^{=2}(p)$.*

*(2) If $k_1 > k_2$, $U_p$ acts locally finitely on $H^0(X_1^{=2}(p), \omega(k_1, k_2, k_3))$. Let $f_{\mathrm{Kl}}$ be the ordinary projector associated with $U_p$.*

*(3) For $k_1 > k_2$ and $k_1 + k_3 > 1$, $\mathrm{tr} \circ U_p^n \circ \mathrm{res} = (T_p^{1,1})^n$ as elements of $\mathrm{End}(e_{\mathrm{Kl}} H^0(\mathfrak{X}_1^{=2}, \omega(k_1, k_2, k_3)))$. Consequently, the composition*

$$e_{\mathrm{Kl}} H^0(X_1^{=2}, \omega(k_1, k_2, k_3)) \xrightarrow{\mathrm{res}} e_{\mathrm{Kl}} H^0(X_1^{=2}(p), \omega(k_1, k_2, k_3)) \xrightarrow{U_p} f_{\mathrm{Kl}} H^0(X_1^{=2}(p), \omega(k_1, k_2, k_3)),$$

*is a bijection.*

*(4) Over the mod $p$ (multiplicative) rank $1$ locus, $X_1^{=1} \cong X_1^{=1}(p)$ and $T_p = U_p$.*

*(5) If $k_1 > k_2$ and $k_1 + k_3 > 3$, then each of the two morphisms*

$$e_{\mathrm{Kl}} R\Gamma(X_1, \omega(k_1, k_2, k_3)) \to e_{\mathrm{Kl}} R\Gamma(X_1^{\geq 1}, \omega(k_1, k_2, k_3)) \to f_{\mathrm{Kl}} R\Gamma(X_1^{\geq 1}(p), \omega(k_1, k_2, k_3)),$$

*are all quasi-isomorphisms.*

3.1.5. *Integral control theorem.* Now we come to the main upshot.

**Theorem 3.9** (Comparison of integral ordinary cohomology; [Ng, Theorem 13.0.1])**.** *For $k_1 > k_2$ and $k_1 + k_3 > 3$, we have quasi-isomorphisms*

$$e_{\mathrm{Kl}} R\Gamma(\mathfrak{X}, \omega(k_1, k_2, k_3)) \xrightarrow{\sim} e_{\mathrm{Kl}} R\Gamma(\mathfrak{X}^{\geq 1}, \omega(k_1, k_2, k_3)) \xrightarrow{\sim} f_{\mathrm{Kl}} R\Gamma(\mathfrak{X}^{\geq 1}(p), \omega(k_1, k_2, k_3)).$$

As mentioned above, the integral control theorem comes right away as we know much more about the $T_p$-correspondence.

**Definition 3.10.** Let the Hida complex $M_{\kappa, k_2, k_3}$ be a $\Lambda$-module defined as

$$M^{\bullet}_{\kappa, k_2, k_3} = f_{\mathrm{Kl}} R\Gamma(\mathfrak{X}^{\geq 1}(p), \mathfrak{F}^{\kappa}).$$

**Theorem 3.11** (Integral control theorem; [Ng, Theorem B])**.** *The Hida complex $M^{\bullet}_{\kappa, k_2, k_3}$ is a perfect complex of $\Lambda$-modules concetnrated in degrees $[0, 1]$. For any $k_1 > k_2$, $k_1 + k_3 > 3$, we have an integral control theorem*

$$M^{\bullet}_{\kappa, k_2, k_3} \otimes^L_{\Lambda, k_1} \mathbb{Z}_p \cong e_{\mathrm{Kl}} R\Gamma(X, \omega(k_1, k_2, k_3)).$$

**Remark 3.12.** The $\mathbb{Q}_p$-contorl theorem would have more relaxed regularity condition, including limit of discrete series. This however requires invoking the higher Coleman theory and in particular the computation of Hecke eigenvalues.

3.2. **Concentration in degree** $1$**.** We complement the higher Hida theory of [Ng] with a vanishing theorem.

**Proposition 3.13.** *If $k_1 + k_3 > 3$ and $k_2 + k_3 < 0$, the perfect complex*

$$M^{\bullet}_{\kappa, k_2, k_3} \otimes^L_{\Lambda, k_1} \mathbb{Z}_p \cong e_{\mathrm{Kl}} R\Gamma(X, \omega(k_1, k_2, k_3)),$$

*is concentrated in degree $1$.*

*Proof.* We know the characteristic zero classes

$$e_{\mathrm{Kl}} R\Gamma(X, \omega(k_1, k_2, k_3)) \otimes^L_{\mathbb{Z}_p} \mathbb{Q}_p,$$

is concentrated in degree 1 by the Lie algebra cohomology computation. Thus, we only need to show that

$$e_{\mathrm{Kl}} H^0(X_1, \omega(k_1, k_2, k_3)) = 0,$$

which by Theorem 3.8(5) is isomorphic to $f_{\mathrm{Kl}} H^0(X_1^{\geq 1}(p), \omega(k_1, k_2, k_3))$.

To show it, we closely follow the proof of [LPSZ, Proposition 3.10]. Let $C \subset \mathcal{A}[\pi]$ be the universal multiplicative subgroup of order $p$. Consider

$$\omega(k_1, k_2, k_3) = \mathrm{Sym}^{k_1 - k_2} \omega_\sigma \otimes \omega_{\overline{\sigma}}^{k_2 + k_3} \twoheadrightarrow \omega_C^{k_1 - k_2} \otimes \omega_{\overline{\sigma}}^{k_2 + k_3}.$$

As $U_p$ is a multiple of $p$ as an endomorphism of its kernel sheaf, we know that the map

$$f_{\mathrm{Kl}} H^0(X_1^{\geq 1}(p), \omega(k_1, k_2, k_3)) \to f_{\mathrm{Kl}} H^0(X_1^{\geq 1}(p), \omega_C^{k_1 - k_2} \otimes \omega_{\overline{\sigma}}^{k_2 + k_3}),$$

is injective. Thus, it is sufficient to show that $f_{\mathrm{Kl}} H^0(X_1^{\geq 1}(p), \omega_C^{k_1 - k_2} \otimes \omega_{\overline{\sigma}}^{k_2 + k_3}) = 0$.

Let $E_0$ be a CM ordinary elliptic curve defined over a field $k$ of characteristic $p$ which has a CM by $\mathcal{O}_K$. Choose a multiplicative subgroup $C \subset E_0$ and the appropriate level structure away from $p$. Then, there is a natural embedding

$$j : X_H(\mathcal{P}_H)_1 \times_{\mathbb{F}_p} k \to X_1^{\geq 1}(p),$$

(see §2.4), sending $(A', C') \mapsto (A' \times E_0, C' \times C)$; here the $p$-level $\mathcal{P}_H$ corresponds to parametrizing $C' \subset A'[\pi]$ of order $p$. As $j^* \omega_H = \mathcal{O}$, for any $a \in H^0(X_1^{\geq 1}(p), \omega_H^{k_1 - k_2} \otimes \omega_{\overline{\sigma}}^{k_2 + k_3})$, $j^* a$ is a mod $p$ holomorphic automorphic form for $\mathrm{U}(1,1)$ with negative weight, which must be zero. The same applies for any prime-to-$p$ Hecke translate of the embedded modular curves. It is thus sufficient to prove that the set of prime-to-$p$ Hecke translates of points of form $A' \times E_0$ forms a Zariski dense subset of $X_1^{\mathrm{ord}}$, using the irreducibility of $X_1^{\mathrm{ord}}(p) \to X_1^{\mathrm{ord}}$. Suppose the contrary; the set then misses a Zariski open subset of $X_1^{\mathrm{ord}}$. Thus, the set is contained in a finite union of curves. This finite union of curves is invariant under all prime-to-$p$ Hecke correspondences, so this violates the fact that the $\ell$-adic monodromy of Hecke-invariant subvariety must contain the derived group of $G$ (cf. [Ch]). □

3.3. **Representation theory of the ordinary part.** Although many parts of higher Hida theory as in [Pi] is established in [Ng], the bridge between cohomology and automorphic representations is not explored. To analyze it, we need to compute the Hecke eigenvalues of certain local representations of $G(K_v)$. Fortunately, $G(K_v) \cong \mathrm{GL}_3(K_v) \times K_v^\times$.

**Lemma 3.14.** *The spherical Hecke algebra*

$$\mathcal{H}_v := C_c(G(K_v)//\mathcal{G}(\mathcal{O}_v), \mathbb{Z}),$$

*is isomorphic to* $\mathbb{Z}[T_{v,0}, T_{v,0}^{-1}, T_{v,1}, T_{v,2}, T_{v,3}, T_{v,3}^{-1}]$, *where*

$$T_{v,0} = \mathcal{G}(\mathcal{O}_v)(1,p)\mathcal{G}(\mathcal{O}_v), \quad T_{v,1} = \mathcal{G}(\mathcal{O}_v)(\mathrm{diag}(p,1,1),1)\mathcal{G}(\mathcal{O}_v),$$

$$T_{v,2} = \mathcal{G}(\mathcal{O}_v)(\mathrm{diag}(p,p,1),1)\mathcal{G}(\mathcal{O}_v), \quad T_{v,3} = \mathcal{G}(\mathcal{O}_v)(\mathrm{diag}(p,p,p),1)\mathcal{G}(\mathcal{O}_v),$$

*under the identification* $G(K_v) \cong \mathrm{GL}_3(K_v) \times \mathbb{G}_m(K_v)$. *The Hecke polynomial is*

$$Q_v(X) = 1 - T_{v,1}X + p^3 T_{v,2} X^2 - p^6 T_{v,3} X^3.$$

Similarly, let $\mathcal{H}_{\mathcal{P}}^{+}$ be the subalgebra of the "mirahoric" Hecke algebra $C_c(G(K_v)//\mathcal{P},\mathbb{Z})$ generated by

$$U_{v,0}=\mathcal{P}(1,p)\mathcal{P},\quad U_{v,1}=\mathcal{P}(\mathrm{diag}(p,1,1),1)\mathcal{P},$$
$$U_{v,2}=\mathcal{P}(\mathrm{diag}(p,p,1),1)\mathcal{P},\quad U_{v,3}=\mathcal{P}(\mathrm{diag}(p,p,p),1)\mathcal{P},$$

under the identification $G(K_v)\cong \mathrm{GL}_3(K_v)\times\mathbb{G}_m(K_v)$. This is identified with the center of the mirahoric Hecke algebra, and is naturally isomorphic to the spherical Hecke algebra.

Now assume that $\pi_v$ is an irreducible complex smooth admissible represntation of $G(K_v)$ which is spherical, namely $\pi_v^{\mathcal{G}(\mathcal{O}_v)}\neq 0$. Let $\theta_{\pi_v}:\mathcal{H}_v\to\mathbb{C}$ be the Hecke eigensystem, and let $(\alpha,\beta,\gamma)$ be the reciprocal of the roots of $\theta_{\pi_v}(Q_v(X))$, ordered in a way that $|\alpha|_p\geq|\beta|_p\geq|\gamma|_p$. By the very definition, we have the following

**Lemma 3.15.** *The eigenvalues for $T_{v,1},T_{v,2}$ and $T_{v,3}$ acting on $\pi_v^{G(\mathcal{O}_v)}$ are, respectively,*

$$\alpha+\beta+\gamma,\quad p^{-3}(\alpha\beta+\beta\gamma+\gamma\alpha),\quad p^{-6}\alpha\beta\gamma.$$

**Lemma 3.16.** *If $\pi_v$ is furthermore generic (namely, an irreducible spherical principal series), then $\pi_v^{\mathcal{P}}$ is three-dimensional, and the eigenvalues of $U_{v,1},U_{v,2},U_{v,3}$ on $\pi_v^{\mathcal{P}}$ are, respectively,*

$$U_{v,1}:\quad \alpha,\beta,\gamma,$$
$$U_{v,2}:\quad p^{-3}\alpha\beta,p^{-3}\beta\gamma,p^{-3}\gamma\alpha,$$
$$U_{v,3}:\quad p^{-6}\alpha\beta\gamma,p^{-6}\alpha\beta\gamma,p^{-6}\alpha\beta\gamma.$$

*Proof.* This follows from the computation of [GT, Cor. 3.2.2] adapted to the Satake transform of [Re, Theorem 2]. □

From this, the following is immediate.

**Corollary 3.17.** *Assume that the set $\{\alpha\beta,\beta\gamma,\gamma\alpha\}$ has 3 distinct elements. Then, the map*

$$\pi_v^{\mathcal{G}(\mathcal{O}_v)}\hookrightarrow\pi_v^{\mathcal{P}}\twoheadrightarrow(\pi_v^{\mathcal{P}})_{\alpha\beta},$$

*is an isomorphism, where $(\pi_v^{\mathcal{P}})_{\alpha\beta}$ is the $p^{-3}\alpha\beta$-eigenspace in $\pi_v^{\mathcal{P}}$ for $U_{v,2}$.*

For example, such condition happens for any Klingen-ordinary representation appearing in $\varinjlim_{K_p}H^1(X(K^pK_p)_{\overline{\mathbb{Q}}_p},\omega(k_1,k_2,k_3)(-D))$ with $k_1+k_3>3$, $k_2+k_3<-1$.

## 4. The $p$-adic pushforward map

We construct the $p$-adic pushforward map between (higher) Hida theories of $H$ and $G$, as done in [LPSZ, §4]. In this section, we fix a tame level $K^p\subset G(\mathbb{A}_f)$. A key point here is an embedding of Igusa towers.

**Proposition 4.1.** *We have*

$$H(K_v)\cap\gamma\,\mathrm{Kl}(p^m)\gamma^{-1}=\Gamma_0(p^m),$$

*where $\gamma=\begin{pmatrix}&&1\\&1&\\1&&\end{pmatrix}$, $\Gamma_0(p^m)\subset H(\mathcal{O}_v)$ is the preimage of $B_H(\mathbb{Z}/p^m\mathbb{Z})\subset H(\mathbb{Z}/p^m\mathbb{Z})$ via the reduction map $H(\mathcal{O}_v)\twoheadrightarrow H(\mathbb{Z}/p^m\mathbb{Z})$.*

Thus, as far as the restriction is concerned, the corresponding tame level for $H$ would be $K_H^p:=H(\mathbb{A}_f)\cap K^p$, which we also omit from the notations alonside with $K^p$.

4.1. **Partial Igusa tower and Hida theory for $H$.** Now we define a partial Igusa tower for $H$ to be embedded into $\mathfrak{IG}_G^{\geq 1}(p^m)$. Recall that $H = \mathrm{GU}(1,1) \times_{\mathrm{GL}_1} \mathrm{GU}(1)$, so it admits a projection map $p : H \to \mathrm{GU}(1,1)$ from which there are natural maps between the corresponding Shimura varieties. Let $X_{\mathrm{GU}(1,1)}$ be the[1] compactified Shimura variety for $\mathrm{GU}(1,1)$ with level $K^p_{\mathrm{GU}(1,1)}\,\mathrm{GU}(1,1)(\mathbb{Z}_p)$. Let $\mathfrak{X}_{\mathrm{GU}(1,1)}/\operatorname{Spf}_{\mathbb{Z}_p}$ be the $p$-adic formal completion of $X_{\mathrm{GU}(1,1)}$, and $\mathfrak{X}^{\mathrm{ord}}_{\mathrm{GU}(1,1)}$ be the ordinary locus of $\mathfrak{X}_{\mathrm{GU}(1,1)}$. Over $\mathfrak{X}^{\mathrm{ord}}_{\mathrm{GU}(1,1)}$, there is an Igusa tower $\mathfrak{IG}^{\mathrm{ord}}_{\mathrm{GU}(1,1)}(p^m)$, which is a moduli space of embeddings $\alpha : \mu_{p^m} \hookrightarrow \mathcal{A}_{\mathrm{GU}(1,1)}[\pi]$, where $\mathcal{A}_{\mathrm{GU}(1,1)}$ is the universal semi-abelian scheme over $\mathfrak{X}_{\mathrm{GU}(1,1)}$.

**Definition 4.2.** Over $\mathfrak{X}^{\mathrm{ord}}_H$, we define the partial Igusa tower $\mathfrak{IG}^{\mathrm{ord}}_H(p^m)$ to be the pullback

$$\mathfrak{IG}^{\mathrm{ord}}_H(p^m) := p^*\mathfrak{IG}^{\mathrm{ord}}_{\mathrm{GU}(1,1)}(p^m).$$

**Remark 4.3.** The way the Igusa tower is defined here is slightly different from more widely used construction of Igusa tower in the construction of $p$-adic $L$-function for unitary groups, as in [EHLS], [Ha], [HLS]. Namely, the usual way it was defined is to take the fibre product of the two ordinary Igusa towers of $\mathrm{GU}(1,1)$ and $\mathrm{GU}(1)$ over their respective ordinary loci. The objective of the prior works is to compare Hida theory ($H^0$) of the smaller group with the Hida theory (again $H^0$) of the bigger group, whereas our objective is to compare Hida theory of the smaller group with the *higher Hida theory* ($H^1$) of the bigger group. Accordingly, the partial Igusa tower over $H$ should naturally embed into the partial Igusa tower $\mathfrak{IG}_G^{\geq 1}(p^m)$ over $G$. We are in a fortunate situation where the complementary part of $\mathrm{GU}(1)$ has no meaningful geometric structure, so that the situation is not too different from the case of prior works.

From this, we can define the Hida theory for $\mathrm{GU}(1,1)$ and $H$.

**Definition 4.4.** Let $\pi : \mathfrak{IG}^{\mathrm{ord}}_{\mathrm{GU}(1,1)}(p^\infty) \to \mathfrak{X}^{\mathrm{ord}}_{\mathrm{GU}(1,1)}(p)$ be the projection map. Let

$$\mathfrak{F}^{\kappa}_{\mathrm{GU}(1,1)} = (\pi_*\mathcal{O}_{\mathfrak{IG}^{\mathrm{ord}}_{\mathrm{GU}(1,1)}(p^\infty)}\widehat{\otimes}_{\mathbb{Z}_p}\Lambda)[\Gamma = \kappa - k_2] \otimes W_{\mathrm{GU}(1,1)}(k_2, k_2; -2),$$

be the sheaf on $\mathfrak{X}^{\mathrm{ord}}_{\mathrm{GU}(1,1)}(p)$. Given any character $\lambda : \mathbb{Z}_p^\times \to R^\times$ for a $p$-adic ring $R$, we define $\mathfrak{F}^{\lambda}_{\mathrm{GU}(1,1)} := \mathfrak{F}^{\kappa}_{\mathrm{GU}(1,1)}\widehat{\otimes}_{\Lambda,\lambda}R$. Let the space of $p$-adic modular forms for $\mathrm{GU}(1,1)$ be defined as

$$\mathcal{M}^{(\kappa_1,\kappa_2)}_{\mathrm{GU}(1,1)}(R) = H^0(\mathfrak{X}^{\mathrm{ord}}_{\mathrm{GU}(1,1)}(p), \mathfrak{F}^{\kappa_1}_{\mathrm{GU}(1,1)})^{(\mathbb{Z}/p\mathbb{Z})^\times},$$

and similarly defined is the space of cusp forms. Similarlry, the space of $p$-adic modular forms for $H$ is defined as

$$\mathcal{M}^{(\kappa_1,\kappa_2)}_{H}(R) = H^0(\mathfrak{X}^{\mathrm{ord}}_{H}(p), \mathfrak{F}^{\kappa_1}_{H})^{(\mathbb{Z}/p\mathbb{Z})^\times},$$

and similarly defined is the space of cusp forms.

4.2. **Maps between Igusa towers.**

[1]There is only one choice of compactification, as $Y_{\mathrm{GU}(1,1)}$ is a curve.

**Definition 4.5.** We define the tower of schemes

$$\begin{tikzcd}
\mathfrak{IG}_H^{\mathrm{ord}}(p^m) \arrow[r, "\widetilde{\iota}_m"] \arrow[d] & \mathfrak{IG}_G^{\geq 1}(p^m) \arrow[d] \\
\mathfrak{X}_H^{\mathrm{ord}}(p^m) \arrow[r, "\iota_m"] \arrow[d] & \mathfrak{X}_G^{\geq 1}(p^m) \arrow[d] \\
\mathfrak{X}_H \arrow[r, "\iota"] & \mathfrak{X}_G
\end{tikzcd}$$

as follows.

- $\iota$ is induced from the inclusion of Shimura data, induced from $H \hookrightarrow G$;
- $\iota_m$ sends $((A_1, C), A_2)$ to $(A_1 \times A_2, C)$, where $A_1$ and $A_2$ are CM abelian schemes of dimension 2 and 1, respeectively, and $C \subset A_1[\pi]$ is a multiplicative subgorup of order $p$.
- $\widetilde{\iota}_m$ sends $\alpha : \mu_{p^m} \cong C \subset A_1[\pi]$ to $\alpha : \mu_{p^m} \hookrightarrow (A_1 \times A_2)[\pi]$.

**Proposition 4.6.** *The above diagram is consisted of two Cartesian squares. In particular, $\iota_m$ and $\widetilde{\iota}_m$ are closed embeddings.*

*Proof.* The same proof works, *verbatim*, as in [LPSZ, Proposition 4.6, Lemma 4.7]. □

This results in the following

**Proposition 4.7.** *For any integers $k_1 \geq \kappa_1 \geq k_2 \geq \kappa_2 \geq -k_3$, we have the following diagram of sheaves on $\mathfrak{X}_H^{\mathrm{ord}}(p)$,*

$$\begin{tikzcd}
\omega_H(\kappa_1, \kappa_2; -4) \arrow[r, hook] \arrow[d] & \iota_1^* \omega_G(k_1, k_2, k_3) \arrow[d] \\
\mathfrak{F}_H^{(\kappa_1, \kappa_2)} \otimes \omega_H(0, 0; -4) \arrow[r, hook] & \iota_1^* \mathfrak{F}_G^{(k_1, k_2, k_3)}
\end{tikzcd}$$

From this, we can apply the pushforward to $\mathfrak{F}_G^{\kappa}$ to obtain

$$\mathfrak{F}_H^{(\kappa_1 - 1, \kappa_2 + 2)} \otimes \omega_H(0, 0; -4) \to \iota_1^*(\mathfrak{F}_G^{\kappa}),$$

where $(-1, 2)$ came from the conormal bundle $\alpha_{G/H}$. This yields a $p$-adic pushforward

$$\iota_* : \mathcal{S}_{(\kappa_1, \kappa_2)}(R) \to H^1(\mathfrak{X}_G^{\geq 1}(p), \mathfrak{F}_G^{\kappa}(-D_G)),$$

which obviously extends the pushforward of coherent cohomology.

## 5. Integral representations of $L$-functions

We use the integral representation appearing in [GPS] for the standard $L$-function of generic automorphic representation of $\mathrm{GU}(2, 1)$. We extend the strategy to relate the GGP period integral for $\mathrm{GU}(1, 1) \subset \mathrm{GU}(2, 1)$ to the standard $L$-function for $\mathrm{GU}(2, 1) \times \mathrm{GU}(1, 1)$, which we develop here. This unfolding of integral formula is independent of the recent fascinating development of resolution of Ichino–Ikeda conjecture ([Zh], [BPC+], [BPL+]) and sheds new light on the period integral. This uses the *generalized Whittaker model*[2] due to a Japanese school of mathematicians, most notably Shintani, Yamashita and Murase–Sugano.

[2]There are unfortunately no concencus on what generalized Whittaker models mean across the literature. In particular, in some literature, generalized Whittaker models are special cases of degenerate Whittaker models in the sense of Moeglin–Waldspurger. Our meaning of generalized Whittaker model is different from this. Another

5.1. **Gelbart–Piatetski-Shapiro integral for** $\mathrm{GU}(2,1)$**.** Motivated by [GPS], we are led to consider the zeta integral as follows.

**Definition 5.1.** Let $\Pi$ be a cuspidal automorphic representation of $\mathrm{GU}(2,1)(\mathbb{A}_{\mathbb{Q}})$, and $f \in \Pi$. Let $\mu$ be an idele class character of $K^\times \backslash \mathbb{A}_K^\times$. Let $\Phi$ be a Schwartz–Bruhat function on $V_H(\mathbb{A}_K)$. Then,

$$Z(f,\mu,\Phi,s) := \int_{Z_G(\mathbb{A})H(\mathbb{Q})\backslash H(\mathbb{A})} f(h)E^\Phi(h;\mu,s)dh,$$

where

$$E^\Phi(h;\mu,s) := \sum_{\gamma \in B_H(\mathbb{Q})\backslash H(\mathbb{Q})} f^\Phi(\gamma h;\mu,s),$$

$$f^\Phi(h;\mu,s) := \int_{\mathbb{A}_K^\times} (h\cdot\Phi)(ty)\mu(t)|t|_K^s d^\times t \qquad \textit{(“Mellin transform”).}$$

By unfolding, one gets a factorizable integral representation.

**Proposition 5.2** ([GPS, Proposition 3.6])**.** *Suppose $\Phi = \prod_w \Phi_w$ is factorizable. Then*

$$\begin{aligned} Z(f,\mu,\Phi,s) &= \int_{Z_G(\mathbb{A})Z_{N_{B_G}}(\mathbb{A})\backslash H(\mathbb{A})} W_f^\psi(h) f^\Phi(h;\mu,s)dh \\ &= \prod_v \int_{Z_G(\mathbb{Q}_v)Z_{N_{B_G}}(\mathbb{Q}_v)\backslash H(\mathbb{Q}_v)} W_v(h) f_v^\Phi(h;\mu,s)dh =: \prod_v Z_v(f,\mu_v,\Phi_v,s), \end{aligned}$$

*where*

- *$Z_{N_{B_G}}$ is the center of the unipotent radical of $B_G$, so*

$$N_{B_G} = \left\{ \begin{pmatrix} 1 & -\overline{a}\theta & z \\ & 1 & a \\ & & 1 \end{pmatrix} \mid z - \overline{z} + \theta|a|^2 = 0 \right\},$$

$$Z_{N_{B_G}} = \left\{ \begin{pmatrix} 1 & & z \\ & 1 & \\ & & 1 \end{pmatrix} \mid z \in \mathbb{Q} \right\},$$

- *$\psi$ is a nontrivial additive character of $\mathbb{Q}\backslash\mathbb{A}_{\mathbb{Q}}$, and $\psi_{N_{B_G}}$ is the character of $N_{B_G}$ defined by $\left(\begin{smallmatrix} 1 & -\overline{a}\theta & z \\ & 1 & a \\ & & 1 \end{smallmatrix}\right) \mapsto \psi(\operatorname{Re} a)$. Here $\operatorname{Re} a = \frac{a+\overline{a}}{2}$.*
- *$W_f^\psi(g) = \int_{N_{B_G}(\mathbb{Q})\backslash N_{B_G}(\mathbb{A}_{\mathbb{Q}})} f(ng)\overline{\psi_{N_{B_G}}(n)}dn$, which thus satisfies that $W_f^\psi(ng) = \psi_{N_{B_G}}(n)W_f^\psi(g)$ for $n \in N_{B_G}(\mathbb{A}_{\mathbb{Q}})$.*
- *$W_v(h)$ is in the local Whittaker model $W(\pi_v,\psi_v) = \{W_v : G(\mathbb{Q}_v) \to \mathbb{C} \mid W_v(ng) = \psi_v(n)W_v(g) \forall n \in N_{B_G}(\mathbb{Q}_v)\}$, which is at most one-dimensional,*
- *$f_v^\Phi(h;\mu,s) = \prod_{w|v} \int_{K_w^\times} (h\cdot\Phi_w)(ty)\mu_w(t)|t|_w^s d^\times t$.*

name for the generalized Whittaker model in our sense that appears in the literature is the *“reduced generalized Gelfand–Graev representation (RGGGR)”*.

5.2. **Generalized Whittaker model, after Shintani, Yamashita, Murase–Sugano.** If we replace the Eisenstein series on $\mathrm{GU}(1,1)$ with a cusp form, then the unfolding picks up more degenerate Whittaker coefficients, because any character $\chi$ of $B$ restricts to a trivial character on $B_H$ which makes the integral corresponding to the Whittaker coefficient with respect to $\chi$ vanish by the cuspidality of the cusp form.

In this subsection, we introduce the ideas behind the so-called *generalized Whittaker model*, as developed by Shintani, Yamashita and Murase–Sugano. For the sake of introduction, the notations in this section are only used here and not related to those used anywhere else.

We first recall that the Fourier expansion of a modular form, or more generally the Whittaker–Fourier expansion of an automorphic form, are all obtained by seeing a form as a function over the unipotent radical $N$ of a parabolic subgroup $P$. In this way, one observes that the chosen form is an $N$-intertwiner, from the (restriction to $N$ of the) automorphic representation that the form generates, to the regular representation $L^2(N_{\mathbb{Q}}\backslash N_{\mathbb{A}})$. This is decomposed into the (maybe) continuous sum of unitary irreducible $N$-representations, and each Fourier coefficient is an intertwiner between the automorphic representation $\pi$ and each irreducible $N$-constituent $\rho$, or by Frobenius reciprocity, $\mathrm{Hom}_G(\pi, \mathrm{Ind}_N^G \rho)$.

The Whittaker–Fourier coefficient corresponds to the case where $N$ is the maximal unipotent subgroup and the irreducible $N$-representations are characters. However, there are many other choices of $N$, and also a unitary irreducible representation of $N$. The main reason why such general theory is ill-developed is because, over local field $F$, $\mathrm{Ind}_{N(F)}^{G(F)} \rho$ is not multiplicity-free, and in most cases not even finite-dimensional. Indeed, if $N$ is smaller than those appearing in the "wave-front set" of $G(F)$-representation $\sigma$, $\mathrm{Hom}_{G(F)}(\sigma, \mathrm{Ind}_N^G \rho)$ is infinite-dimensional (cf. [MW], [GGS])[3]. This prevents one from using this kind of model (at least naively) to unfold Rankin–Selberg integrals.

The idea of the generalized Whittaker model, as developed by Shintani, Yamashita, Murase–Sugano, is to extend the group $N$ to a slightly bigger group $N \subset R \subset P$ and consider $R$-intertwiners. Over a local field, the group $R$ is defined as follows. Let $P = L \ltimes N$ be the Levi decomposition. Let $S \subset L$ be the centralizer of $[\rho] \in \widehat{N}$, the class of $\rho$ in the unitary dual of $N$. Then, one defines $R := S \ltimes N$. The $N$-representation $\rho$ can be trivially extended to a $S$-representation, also denoted as $\rho$. Representations of $R$ are of form $\rho \otimes \psi$, where $\rho$ is the extension of an $N$-representation as above, and $\psi \in \widehat{S}$.

**Definition 5.3.** Let $\pi$ be a $G$-representation. The *generalized Whittaker model* of $\pi$ with respect to $(N, \rho, \psi)$ is the intertwiner

$$\mathrm{Hom}_R(\pi|_R, \rho \otimes \psi) = \mathrm{Hom}_G(\pi, \mathrm{Ind}_R^G(\rho \otimes \psi)).$$

The induced representation $\pi_{\rho,\psi} := \mathrm{Ind}_R^G(\rho \otimes \psi)$ is called a *reduced generalized Gelfand–Graev representation (RGGGR).*

Surprisingly, the existing works suggest that these models have finite multiplicity property, and even multiplicity-free property in favorable situations.

**Slogan.** The RGGGR $\pi_{\rho,\psi}$ has multiplicity one property, namely $\dim_G(\pi, \pi_{\rho,\psi})^\circ \leq 1$.

[3]We will call this often-infinite-dimensional model a *degenerate Whittaker model*, to distinguish it from the generalized Whittaker model we define in this section.

The superscript $\circ$ indicates moderate growth condition should be imposed in the archimedean case. Recall that, even in the classical theory of Jacquet, the space of Whittaker functionals is of dimension larger than one; but *there is exactly one dimensional line spanned by Whittaker functionals of moderate growth.*

We exhibit how to use these models arise in some period integrals to show their Eulerianity, via the theory of Fourier expansions along closed subgroups (cf. [Od]), which, formally speaking, is just the decomposition of an $R$-representation into $\eta$-isotypic parts with $\eta$ running over $\eta \in \widehat{R}$.

**Remark 5.4.** One reason behind our interest towards this theory is for the construction of square-root $p$-adic $L$-function for $\mathrm{U}(n,1) \times \mathrm{U}(n-1,1)$. This more general case should use [MS2], [Ts1], [Ts2], [Ts4].

5.3. **Example I: GGP period integral for** $\mathrm{GU}(2,1) \times \mathrm{GU}(1,1)$**.**

**Definition 5.5.** Let $\Pi, \sigma$ be cuspidal automorphic representations of $\mathrm{GU}(2,1)(\mathbb{A}_{\mathbb{Q}})$ and $\mathrm{GU}(1,1)(\mathbb{A}_{\mathbb{Q}})$, respectively. Let $\mu$ be a character of $\mathrm{GU}(1)(\mathbb{A}_{\mathbb{Q}})$. Let $f \in \Pi$, $g \in \sigma$. Then,

$$Z(f,g,\mu,s) := \int_{Z_G(\mathbb{A})H(\mathbb{Q})\backslash H(\mathbb{A})} f(h)g(h)\mu(h)|\det h|^{s-\frac{1}{2}}dh.$$

We will show that this integral of "GGP form" can be unfolded by using the theory of generalized Whittaker models. To unfold, we need to recall some of the machineries used in [GPS]. There, the theory of Fourier–Jacobi expansions is developed (see also [**?**]GR), which uses the unipotent radical $N_{B'_G}$, which is isomorphic to a Heisenberg group. More precisely, for a local field $F$, the irreducible unitary representations of $N_{B'_G}(F)$ are precisely the characters (factors through $N_{B'_G}/Z_{N_{B'_G}}$) and the Weil representations (characterized by the central character). Taking the Fourier expansion of $f$ with respect to the compact abelian group $[Z_{N_{B'_G}}]$, we have an expansion analogous to the Fourier–Jacobi expansion[4],

$$f(h) = f_{00}(h) + \sum_{\psi \neq 1} f^{\psi}(h),$$

where the sum runs over nontrivial characters of $[Z_{N_{B'_G}}]$, $f_{00}(h) = \int_{[Z_{N_{B'_G}}]} f(nh)dn$, and

$$f^{\psi}(h) = \int_{[Z_{N_{B_G}}]} f(zh)\overline{\psi(z)}dz,$$

is the "Fourier–Jacobi coefficient." As noted above, $f^{\psi}$ corresponds to the Weil representation with central character $\psi$, whereas $f_{00}$ corresponds to all the representations with trivial central character. Thus, one can expect that this will further break down into a sum of the Fourier coefficients corresponding to the characters of $[N_{B'_G}]$, the "Whittaker–Fourier coefficients":

$$f_{00}(h) = \sum_{\chi \neq 1} W^f_{\chi}(h), \qquad W^f_{\chi}(h) = \int_{[N_{B_G}]} f(nh)\overline{\chi(n)}dn,$$

[4]This is indeed the classical Fourier–Jacobi expansion if $f$ is a holomorphic automorphic form, because the holomorphic (limit of) discrete series does not admit a Whittaker mordel, so $f_{00}(g)$, the sum of Whittaker–Fourier coefficients, vanishes; on the other hand, $f^{\psi}(g)$ is a classical Fourier–Jacobi coefficient (cf. [Is]).

where $\chi$ runs over all (nontrivial[5]) characters of $[N_{B'_G}]$.

When $g$ was an Eisenstein series, the integral unfolded so that the major term came from $f_{00}$. On the other hand, if we unfold a cuspidal $g$ against $f_{00}$, since $f_{00}$ is left-$Z_{N_{B_G}}(\mathbb{A})$-invariant, the integral vanishes because of the cuspidality of $g$; note $Z_{N_{B_G}} = N_{B_H}$. Thus, in the case of $\mathrm{GU}(2,1) \times \mathrm{GU}(1,1)$, we are led to consider the integral against the Fourier–Jacobi coefficients $f^{\psi}$.

The imminent problem is that the Fourier–Jacobi coefficient $f^{\psi}$ does not have multiplicity free local model. Rather, we would need to use the generalized Whittaker model. If we let $R$ to be the centralizer of $Z_{N_{B'_G}}$ in $B'_G$, then $R = S \ltimes N_{B_G}$, where $S = \mathrm{U}(1)$ is the reductive part. Thus, irreducible unitary representaitons of $R(F)$ are of form $\chi \otimes \omega_{\psi}$, where $\omega_{\psi}$ is the trivial extension of the Weil representation with central character $\psi$, and $\chi$ is a character of $\mathrm{U}(1)$.

In our case, the **Slogan** was already proven in the literature.

**Proposition 5.6.** *Let $F$ be a local field.*

*(1) If $F$ is nonarchimedean, $\mathrm{Ind}_{R(F)}^{G(F)}(\chi \otimes \omega_{\psi})$ is multiplicity-free.*

*(2) If $F$ is archimedean, $\mathrm{Ind}_{R(F)}^{G(F)}(\chi \otimes \omega_{\psi})^{\circ}$ is multiplicity-free.*

*Proof.* For (1), see [GR]. For (2), see [Is]. □

This implies that the degenerate Whittaker model $f^{\psi}$ decomposes into the sum of generalized Whittaker models,

$$f^{\psi}(g) = \sum_{\chi} f^{\psi,\chi}(g),$$

where $\chi$ runs over the unitary characters of $S(\mathbb{A}) = \mathrm{U}(1)(\mathbb{A})$, and $f^{\psi,\chi}(g)$ is a generalized Whittaker function, so that it transforms as $\chi \otimes \omega_{\psi}$ under the action of $R(\mathbb{A})$. By Proposition 5.6, $f^{\psi,\chi}$ is supported by a multiplicity one model. As $\mathrm{U}(1,1)$-automorphic forms are basically elliptic modular forms, we thus have

$$\begin{aligned}
Z(f,g,\mu,1/2) &= \int_{[H]} \sum_{\psi \neq 1, \chi} f^{\psi,\chi}(h) g(h) \mu(h) dh \\
&= \int_{[\mathrm{GU}(1,1)]} \sum_{\psi \neq 1} f^{\psi,\mu^{-1}}(h) g(h) dh \\
&= \int_{[\mathrm{GU}(1,1)]} \left( \int_{N_{B_H}(\mathbb{Q}) \backslash \mathrm{GU}(1,1)(\mathbb{Q})} f^{\psi,\mu^{-1}}(xh) dx \right) g(h) dh \\
&= \int_{N_{B'_H}(\mathbb{Q}) \backslash \mathrm{U}(1,1)(\mathbb{A})} f^{\psi,\mu^{-1}}(h) g(h) dh \\
&= \int_{N_{B'_H}(\mathbb{A}) \backslash \mathrm{U}(1,1)(\mathbb{A})} f^{\psi,\mu^{-1}}(h) \left( \int_{[N_{B'_H}]} g(nh) \psi(n) dn \right) dh \\
&= \int_{Z_H(\mathbb{A}) N_{B_H}(\mathbb{A}) \backslash H(\mathbb{A})} f^{\psi,\mu^{-1}}(h) W_g^{\psi}(h) dh,
\end{aligned}$$

[5]The coefficient against the trivial character vanishes, because $f$ is cuspidal. One can see that the (non)cuspidality also plays a role in this hierarchy of various depths of generalized Whittaker models.

where $W_g^{\psi}(h)$ is the Whittaker function for $g$ with respect to $\psi$. Note that we ignored the power of $|\det|$ to unclog the formulae, but one can verbatim follow the same process with arbitrary choice of $s$. Now the last integral we yield is manifestly Eulerian due to the *multiplicity one properties of (generalized) Whittaker models*, and we have shown the following

**Proposition 5.7.** *We have*

$$\begin{aligned} Z(f,g,\mu,s) &= \int_{Z_H(\mathbb{A})N_{B_H}(\mathbb{A})\backslash H(\mathbb{A})} f^{\psi,\mu^{-1}}(h)W_g^{\psi}(h)|\det h|^{s-\frac{1}{2}}dh \\ &= \prod_v \int_{Z_H(\mathbb{Q}_v)N_{B_H}(\mathbb{Q}_v)\backslash H(\mathbb{Q}_v)} W_{f,v}^{\psi_v,\mu_v^{-1}}(h)W_{g,v}^{\psi_v}(h)|\det h|^{s-\frac{1}{2}}dh =: \prod_v Z_v(f,g,\mu_v,s), \end{aligned}$$

*where $W_{f,v}^{\psi_v,\mu_v^{-1}}(h)$ and $W_{g,v}^{\psi}(h)$ belong to the generalized Whittaker model for $\Pi_v$ with respect to $\mu_v^{-1}\otimes\omega_{\psi_v}$ and the Whittaker model for $\sigma_v$ with respect to $\psi_v$, respectively.*

5.4. **Example II: Kohnen–Skoruppa integral.** We suggest that the generalized Whittaker model can also explain the so-called *Rankin–Selberg integrals with non-unique local models*. These are Rankin–Selberg integrals that are shown to represent an $L$-function not just by unfolding but also using some auxiliary argument (usually involving Hecke algebra). It is called to have a "non-unique" model because the naive approach unfolds the integral into a degenerate Whittaker model (see §5.2), which is infinite-dimensional as we exactly noted above. We show that, by considering the action of a part of the reductive part as in generalized Whittaker models, a formal manipulation can prove the Eulerianity of the Kohnen–Skoruppa integral[6], a Rankin–Selberg integral with non-unique local models for the spin $L$-function of $\mathrm{Sp}_4$. The following material in this subsection is purely formal, and there are some analytic difficulties to be overcome.

The Kohnen–Skoruppa integral is recast in modern language by [PS]. We freely use the notations of *op. cit.*, although we would have to conjugate the embedding $\mathrm{GL}_{2,L}^* \hookrightarrow \mathrm{GSp}_4$ slightly. For simplicity, we use the split case, $H := \mathrm{GL}_2 \times_{\mathrm{GL}_1} \mathrm{GL}_2 \hookrightarrow G := \mathrm{GSp}_4$, given by

$$\left(\begin{pmatrix} a & b \\ c & d \end{pmatrix}, \begin{pmatrix} a' & b' \\ c' & d' \end{pmatrix}\right) \mapsto \begin{pmatrix} a & & b & \\ & a' & & b' \\ c & & d & \\ & c' & & d' \end{pmatrix}.$$

The integral we consider is

$$J(\phi,s) = \int_{Z(\mathbb{A})H(\mathbb{Q})\backslash H(\mathbb{A})} E(g,s,\Phi)\phi(g)dg,$$

where $\phi$ is a cusp form of a cuspidal automorphic representation $\pi$ on $\mathrm{GSp}_4$, $E(g,s,\Phi)$ is the Klingen Eisenstein series, and $Z$ is the center. This unfolds into

$$J(\phi,s) = \int_{B'(\mathbb{Q})Z(\mathbb{A})\backslash \mathrm{GL}_{2,L}^*(\mathbb{A})} f^{\Phi}(g,s)\phi(g)dg,$$

[6]A related suggestion was made in [Od], although no further justifications were made. As suggested in *op. cit.*, the formal manipulation in this subsection, if made rigorous, shows that the Kohnen–Skoruppa integral would also give rise to an integral representation of the spin $L$-function for automorphic representations whose infinity type is a *generic discrete series*; so far, the integral representation property is only shown for those belonging to a *(anti-) holomorphic discrete series*.

where $B'(\mathbb{Q}) = Q(\mathbb{Q}) \cap H(\mathbb{Q})$ and $Q(\mathbb{Q}) \subset \mathrm{GSp}_4(\mathbb{Q})$ is the Klingen parabolic, $\begin{pmatrix} * & & * & * \\ * & * & * & * \\ * & & * & * \\ & & & * \end{pmatrix}$. Let $B' = LN$ be its Levi decomposition. Then, $L = \mathrm{GL}_2(\mathbb{Q}) \times T$, where $T = \begin{pmatrix} * & \\ & 1 \end{pmatrix} \in \mathrm{GL}_2(\mathbb{Q})$, and $N$ is abelian, just consisted of $\begin{pmatrix} 1 & & & \\ & 1 & & * \\ & & 1 & \\ & & & 1 \end{pmatrix}$. Since $[N]$ is a part of the region of integral, from the Fourier expansion $\phi(g) = \sum_{\psi \in \widehat{[N]}} \phi_\psi(g)$, only the coefficient for the trivial character $\mathbf{1}$ survives, and we get

$$J(\phi, s) = \int_{L(\mathbb{Q})N(\mathbb{A})Z(\mathbb{A})\backslash H(\mathbb{A})} f^\Phi(g, s)\phi_{\mathbf{1}}(g)dg.$$

Now consider $U$, the unipotent radical of $Q$, the Klingen parabolic. It is also a Heisenberg group, with center $N$. The irreducible representations of $U$ with $N$ acting trivially are precisely characters, so

$$\phi_{\mathbf{1}}(g) = \sum_{\chi \in \widehat{[U/N]}} \phi_\chi(g) = \sum_{\gamma \in T} \phi_{\chi_0}(\gamma g),$$

where subscript means the corresponding character for the Fourier coefficient and $\chi_0$ is any choice of nontrivial character. We thus get

$$J(\phi, s) = \int_{\mathrm{GL}_2(\mathbb{Q})N(\mathbb{A})Z(\mathbb{A})\backslash H(\mathbb{A})} f^\Phi(g, s)\phi_{\chi_0}(g)dg,$$

where $\mathrm{GL}_2(\mathbb{Q})$ in the region is the "first factor." Now $\phi_{\chi_0}(g)$ is also an element of the degenerate Whittaker model with respect to Klingen parabolic, which is typically infinite-dimensional. We can try to use the generalized Whittaker model in this case, noting that $S = \mathrm{GL}_2(\mathbb{Q})$ is the reductive part of the centralizer of $N$:

$$\phi_{\chi_0}(g) = \text{“} \sum_{\sigma \in \widehat{[\mathrm{GL}_{2,\mathbb{Q}}]}} \phi_{\sigma \otimes \chi_0} \text{”}. \tag{1}$$

One big problem is that the unitary dual of $\mathrm{GL}_2(F)$ is much more complicated, involving continuous spectrum. Let us pretend that such sum makes sense as we would want. Then, as the region of integral involves $[\mathrm{GL}_{2,\mathbb{Q}}]$, among $\{\phi_{\sigma \otimes \chi_0}\}_{\sigma \in \widehat{[\mathrm{GL}_{2,\mathbb{Q}}]}}$, only the one corresponding to the trivial representation of $\mathrm{GL}_2$, $\phi_{\mathbf{1} \otimes \chi_0}$, survives:

$$J(\phi, s) = \text{“} \int_{R(\mathbb{A})\backslash H(\mathbb{A})} f^\Phi(g, s)\phi_{\mathbf{1} \otimes \chi_0}(g)dg \text{”}.$$

Now we would win the Eulerianity of the integral if we have multiplicity one property of the generalized Whittaker models associated with $\phi_{\mathbf{1} \otimes \chi_0}(g)$. For non-archimedean local models, this is exactly achieved in Section 3 of [MS1]. For archimedean local models, this is achieved in [Hi], with the usual caveat that the multiplicty one holds when one imposes extra *moderate growth condition*. If one settles the question on how to make the sum (1) rigorous, then, as in Jacquet's classical theory, one should be able to show that, for cusp forms, only archimedean local generalized Whittaker functionals of moderate growth can appear.

## 6. Automorphic cohomology

We start with a globally generic $\Pi$ of $\mathrm{GU}(2,1)(\mathbb{A}_{\mathbb{Q}})$. Then, by Arthur's conjectures, which is verified in this case by Rogawski [Ro], $\Pi$ appears[7] with multiplicity one in the automorphic spectrum.

6.1. **Generic discrete series and its $K$-types.** There are three types of discrete series representation for $\mathrm{SU}(2,1)$: *holomorphic*, *generic* and *antiholomorphic*. For each type, one can parametrize them via the unique coefficient that gives rise to nonzero $(\mathfrak{sp}^-, SK_\infty)$-cohomology; here $\mathfrak{sp}^-$ is the corresponding $\mathfrak{p}^-$ for $\mathfrak{su}(2,1)$, and $SK_\infty$ refers to a maximal compact subgroup of $\mathrm{SU}(2,1)$. The convention we are using for weights, etc., match with [KO], so we will freely borrow results from it.

We are mainly interested in the generic discrete series $\Pi_\infty^{\mathrm{gen}}(m_1,m_2)$, which means that

$$H^1(\mathfrak{su}(2,1), SK_\infty; \Pi_\infty^{\mathrm{gen}}(m_1,m_2)\otimes V_G(m_1,m_2)) \neq 0.$$

Recall that $G$-dominance means $m_1, m_2 \geq 0$. For this to hold, $\chi_{V_G(m_1,m_2)^\sim} = \chi_{\Pi_\infty^{\mathrm{gen}}(m_1,m_2)}$. Here, $V_G(m_1,m_2)^\sim$ is the contragredient of $V_G(m_1,m_2)$, which is $V_G(m_2,m_1)$. Note also that the Harish–Chandra parameter and the Blattner parameter coincide for generic discrete series in this case. Thus, it turns out that $\Pi_\infty^{\mathrm{gen}}(m_1,m_2)$ has minimal $K$-type $W_G(m_1+m_2+2,-m_1-1)$. Furthermore, all $K$-types that appear in $\Pi_\infty^{\mathrm{gen}}(m_1,m_2)$ are of the form $W_G((m_1+m_2+2,-m_1-1)+a(1,1)+b(1,-2))$ for $a,b\geq 0$, and they all appear with multiplicity one (this follows from Blattner's formula; cf. [KO, Proposition 3.1]).

In terms of representaionts of $\mathrm{GU}(2,1)(\mathbb{R})$, we denote the generic discrete series $\Pi_\infty^{\mathrm{gen}}(k_1,k_2,k_3)$ to be the infinity type that appears in the cohomology of $H^1(X_G,\omega_G(k_1,k_2,k_3))$. We assume that the globally generic $\Pi$ has infinity type $\Pi_\infty = \Pi_\infty^{\mathrm{gen}}(k_1,k_2,k_3)$. Let $E$ be the field of definition of $\Pi_f$, and let $H^1(\Pi_f)$ be the $\Pi_f$-isotypical subspace of $H^1(X_{G,E},[L_1](-D))$, where $L_1 = W_G(k_1,k_2,k_3)$.

6.2. **Automorphic and $p$-adic period pairing.** The $p$-adic pushforward map is seen to be compatible with the classical pushforward map

$$\iota_* : H^0(X_{H,\mathbb{Q}},\omega_H(\kappa_1-1,\kappa_2+2)) \to H^1(X_{G,\overline{\mathbb{Q}}},\omega_G(k_1,k_2,k_3)),$$

which after applying the Serre duality, we obtain a bilinear $H(\mathbb{A}_f)$-equivariant classical automorphic period pairing

$$H^0(X_{H,E},\omega_H(\kappa_1-1,\kappa_2+2))\otimes H^1(\Pi_f) \to E\{k_1+k_2-k_3-4\},$$

where $\{m\}$ denotes the twist by a character $\ell\mapsto \ell^m$. This automorphic period pairing has an automorphic interpretation as follows.

**Theorem 6.1** (Harris, Su; see [LPSZ, Theorem 5.6]). *Let $\tau_1$ be the minimal $K$-types of highest weight $(m_1+1,-m_1-m_2-2)$. Then, there is a canonical isomorphism of irreducible $G(\mathbb{A}_f)$-representations*

$$\mathrm{Hom}_K(\tau_1,\Pi)\{\frac{k_1+k_2-k_3-4}{2}\} \xrightarrow{\sim} H^1(\Pi_f)_{\mathbb{C}}.$$

[7] In this case, as the half sum of positive roots is integral, there is no difference between $C$-algebraicity and $L$-algebraicity as in [BG].

*Moreover, if $\eta \in H^1(\Pi_f)_{\mathbb{C}}$, corresponding to some $K$-homomorphism $F_\eta : \tau_1 \to \Pi$, then for any $k_1 \geq \kappa_1 \geq k_2 \geq \kappa_2 \geq -k_3$ and any $f \in M_H^{(\kappa_1-1,\kappa_2+2)}$, we have*

$$\langle \iota_*(f), \eta \rangle = \frac{1}{(2\pi i)^2} \int_{Z_G(\mathbb{A})H(\mathbb{Q})\backslash H(\mathbb{A})} F_\eta(v_{\kappa_1,\kappa_2}) f(h) dh,$$

*where $v_{\kappa_1,\kappa_2}$ is the standard basis vector of $\tau_2$ of weight $(-\kappa_1+1, -\kappa_2-2)$.*

Now we think of $p$-adic period pairing: let $p$ be a prime such that $p = v\overline{v}$ is split in $K$ and $\Pi_v$ is unramified. We have seen in §3.3 that $\Pi_v^{\mathcal{P}}$ is 3-dimensional, and the $U_{v,2}$-operator

$$p^{5-k_2}[\mathcal{P}\,\mathrm{diag}(p,p,1)\mathcal{P}],$$

acts on $\Pi_v^{\mathcal{P}}$ with eigenvalues

$$\left\{ \frac{\alpha\beta}{p^{k_2-2}}, \frac{\beta\gamma}{p^{k_2-2}}, \frac{\gamma\alpha}{p^{k_2-2}} \right\},$$

where $\{\alpha, \beta, \gamma\}$ is the Hecke-eigenvalues of $\Pi_v$. By the standard "$p$-level descent argument" in Hida theory (e.g. [Ti, Lemma 2.21]), we have the following

**Proposition 6.2.** *For $m \geq 1$, we have a factorization of the diagram*

$$\begin{tikzcd}
\Pi_v^{\mathrm{Kl}(p^m)} \arrow[r, "U_{v,2}"] \arrow[d, hook] & \Pi_v^{\mathrm{Kl}(p^m)} \arrow[d, hook] \\
\Pi_v^{\mathrm{Kl}(p^{m+1})} \arrow[r, "U_{v,2}"'] \arrow[ur, dashed, "\xi_m"] & \Pi_v^{\mathrm{Kl}(p^{m+1})}
\end{tikzcd}$$

Using this, we define the following projector.

**Definition 6.3.** Let $\eta \in H^1(\Pi_f)^{\mathrm{Kl}(p)}$. Then, let $\eta_m \in H^1(\Pi_f)^{\mathrm{Kl}(p^m)}$ be characterized as follows.

(1) For $v \in H^1(\Pi_f^\vee)^{\mathrm{Kl}(p^m), U_{v,2}\text{-nilpotent}}$, $\langle v, \eta_m \rangle = 0$.
(2) For $v \in H^1(\Pi_f^\vee)^{\mathrm{Kl}(p)}$, $\langle v, \eta \rangle = \langle \xi_m(v), \eta_m \rangle$, where $\xi_m$ is as defined in Proposition 6.2.

**Proposition 6.4.** *If $\eta_1$ lies in the $U_{v,2}$-eigenspace with eigenvalue $\lambda$, then*

$$\eta_m = \left(\frac{p^{1-k_2}}{\lambda}\right)^{m-1} \sum_{a \in \mathbb{Z}/p^{m-1}\mathbb{Z}} \begin{pmatrix} 1 & & p^m a \\ & 1 & \\ & & 1 \end{pmatrix} \begin{pmatrix} p & & \\ & p & \\ & & 1 \end{pmatrix}^{m-1} \eta.$$

*Proof.* The proof is the same as that of [LPSZ, Proposition 5.9]. One can choose the coset representatives of the $U_{v,2}$-operator on $\mathrm{Kl}(p^m)$-level to be upper triangular and to commute with $\begin{pmatrix} 1 & & \\ p^m & 1 & \\ & & 1 \end{pmatrix}$, and also to be compatible with the normalized trace

$$\eta_m = \frac{1}{p^3} \sum_{\gamma \in \mathrm{Kl}(p^m)/\mathrm{Kl}(p^{m+1})} \gamma \eta_{m+1}.$$

□

We say that $\Pi$ is Klingen-ordinary at $p$ if $\Pi_v^{\mathrm{Kl}(p)}$ has an $U_{v,2}$-eigenvalue $\lambda$ which is a $p$-adic unit. For such $\Pi$, we construct a $p$-adic interpolation of period pairing, and show that it has the desired interpolation property.

**Proposition 6.5** ($p$-adic interpolation of cup product in coherent cohomology)**.** *Let $\eta \in H^2(\Pi_f)$ be a class in the ordinary eigenspace for $U_{v,2}$. Consider $\kappa_1, \kappa_2 : \mathbb{Z}_p^\times \to R^\times$ for a flat $p$-adically complete $\mathcal{O}_{E_{\widetilde{v}}}$-algebra $R$, for $\widetilde{v}|p$, we define, for $\mathcal{E} \in \mathcal{S}_{(\kappa_1,\kappa_2)}(R)$, an element $\langle \iota_*(\mathcal{E}), \eta\rangle \in R[1/p]$, as the image of $\mathcal{E}$ under the composition*

$$\begin{aligned}\mathcal{S}_{(\kappa_1,\kappa_2)}(R) &\xrightarrow[(\S 4.2)]{\iota_*} R\widehat{\otimes}_{\mathbb{Z}_p} H^1(\mathfrak{X}_G^{\geq 1}(p), \mathfrak{F}_G^{(k_1,k_2,k_3)}(-D_G)) \\ &\xrightarrow{f_{\mathrm{Kl}}} R[1/p]\widehat{\otimes}_{\mathbb{Z}_p} f_{\mathrm{Kl}} H^1(\mathfrak{X}^{\geq 1}(p), \mathfrak{F}^{(k_1,k_2,k_3)}(-D_G)) \\ &\xrightarrow{\sim} R[1/p] \otimes_{\mathbb{Q}_p} f_{\mathrm{Kl}} H^1(X_G(p), \omega(k_1,k_2,k_3)) \\ &\xrightarrow{\langle -,\eta\rangle} R[1/p].\end{aligned}$$

*If $\phi : R \to E_{\widetilde{v}}$ is a continuous ring homomorphism, $k_1 > \kappa_1 + 1 > k_2 > \kappa_2 - 2 > -k_3$, and if $\phi(\mathcal{E})$ is the restriction of a classical holomorphic modular form $E_\phi \in M_{(\kappa_1,\kappa_2)}(K_H(p), L)$ to $\mathcal{X}_H^{\mathrm{ord}}$, we have*

$$\phi(\langle \iota_*(\mathcal{E}), \eta\rangle) = \langle \iota_{1,*}(E_\phi), \eta\rangle.$$

*The same holds for a deeper $p$-level $E_\phi$.*

*Proof.* As in the proof of [LPSZ, Proposition 5.12], the only subtlety arises when $E_\phi$ is not cuspidal but nevertheless cuspidal at the ordinary locus. However, a get-around can be given by using the overconvergent cohomology, where there is a classicality theorem [Ng, Theorem 18.4.1, Theorem 18.4.2] for all classes of small slope (which, in particular, include ordinary classes). Since the classical cohomology has the same cuspidal and non-cuspidal ordinary cohomology, we get the desired result anyways. □

6.3. **Nearly sheaves and nearly holomorphic modular forms.** We develop the theory of "nearly" sheaves in the sense of [LPSZ, §6], and connect it with nearly holomorphic modular forms and their incarnation as $p$-adic modular forms. It involves automorphic vector bundles that do not necessarily come from Levi. Namely, the "nearly-ness" comes from considering automorphic vector bundles $[V]$ where $V$ is an algebraic representation of $P_-$ which does not factor through $K$.

**Definition 6.6.** Let $V$ be an algebraic representation of $P_-$. Then, let $\mathrm{Fil}^n$ be the sum of weight spaces of weight $\lambda(k_1,k_2,k_3;c)$ with $k_1+k_2-k_3-c \geq 2n$. This filtration is $P_-$-stable, and the action of $P_-$ on $\mathrm{gr}^n V = \mathrm{Fil}^n V/\mathrm{Fil}^{n+1} V$ factors through $M$.

We apply this to $V_G(k_1,k_2,k_3;-k_1-k_2+k_3)$, where the grading spans for degrees $0 \leq n \leq k_1+k_2-k_3$.

**Definition 6.7.** Let $\widetilde{L}_1$ be the $P_-$-representation

$$\widetilde{L}_1 := \left(\mathrm{Fil}^{\frac{3}{2}(k_2-k_3)} V_G(k_1,k_2,k_3;-k_1-k_2+k_3)\right) \otimes W(1,-1,0;0),$$

and let $\widetilde{L}_1'$ be the top filtration quotient of $\widetilde{L}_1$. We similarly define the dual notion $\widetilde{L^\vee}_1$ and $\widetilde{L^\vee}_1'$ for $L_1^\vee$.

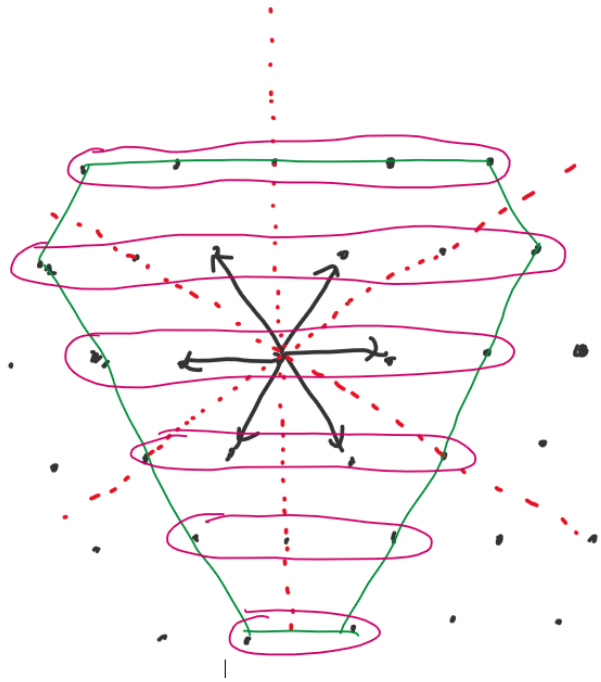
There is a natural inclusion $L_1 \hookrightarrow \widetilde{L}_1'$. As $\Pi_f$ does not contribute to $H^1$ coherent cohomology of other automorphic vector bundle coming from the Levi, we know that

$$H^1(\Pi_f) \to H^1(X_{G,E}, [\widetilde{L}_1'](-D))[\Pi_f] \leftarrow H^1(X_{G,E}, [\widetilde{L}_1](-D))[\Pi_f],$$

are all isomorphisms.

**Definition 6.8.** Given $\eta \in H^1(\Pi_f)$, let $\widetilde{\eta} \in H^1(X_{G,E}, [\widetilde{L}_1](-D))[\Pi_f]$ be the corresponding class under the above isomorphism.

The branching of algebraic representations of $\mathrm{GL}_3$ to $\mathrm{GL}_2$ works pictorially as follows.



By branching law, $V_H := V_H(\kappa_1, \kappa_2)$ is an $H$-subrepresentation if and only if $k_1 \geq \kappa_1 \geq k_2 \geq \kappa_2 \geq -k_3$. From this, we obtain the pushforward of nearly sheaves

$$\iota_*^{\mathrm{nearly}} : H^0(X_{H,\mathbb{Q}}, [V_H] \otimes \omega_H(1, -1, 0; 0)) \to H^1(X_{G,\mathbb{Q}}, [\widetilde{L^\vee}_1]).$$

Using this, we have a cohomological interpretation of period integral for nearly holomorphic modular forms,

$$H^0(X_{H,E}, [V_H] \otimes \omega_H(1, -1, 0; 0)) \otimes H^1(\Pi_f) \to E,$$
$$(f, \eta) \mapsto \langle \iota_*^{\mathrm{nearly}}(f), \widetilde{\eta} \rangle.$$

**Definition 6.9.** Let the *Hodge splitting* be the $C^\infty$-splitting

$$[V_H] \otimes \omega_H(1, -1, 0; 0) \to \omega_H(\kappa_1 + 1, \kappa_2 - 1),$$

which exists as the category of finite dimensional representations of a compact Lie group is semisimple.

Comparing the $K$-types of $\Pi_\infty^{\mathrm{gen}}$ and that of $\widetilde{L}_1$, one can compute the $\Pi_f$-isotypic part of $H^1(X_{G,\mathbb{C}}, [\widetilde{L}_1](-D))$ in terms of Hodge splitting.

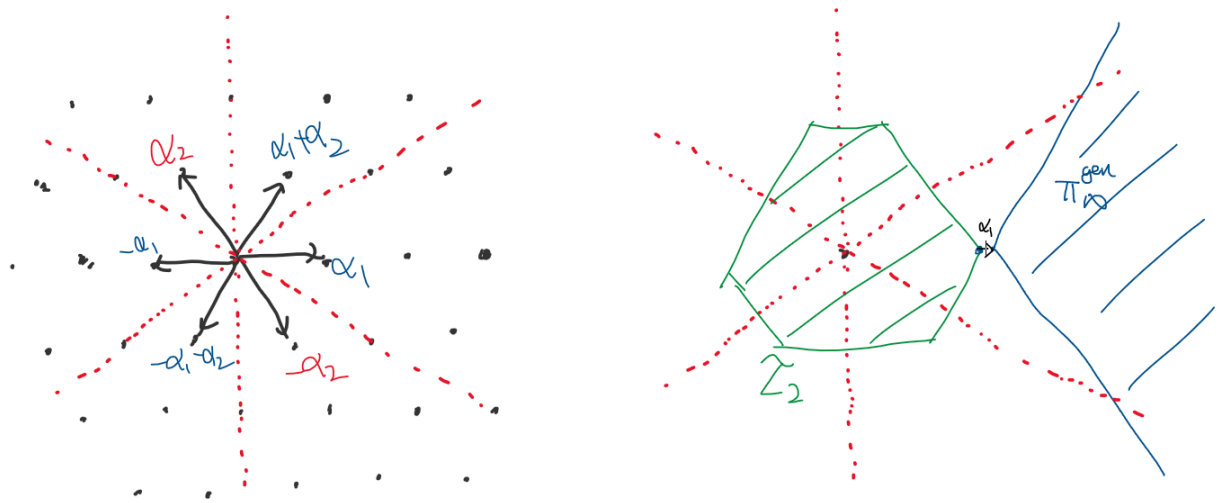
**Theorem 6.10.** *Let $E \in H^0(X_{H,\mathbb{C}}, [V_H] \otimes \omega_H(1, -1, 0; 0))$ and let $E_\infty \in H^0(X_{H,\mathbb{C}}, \omega(\kappa_1 + 1, \kappa_2 - 1)_{C^\infty})$ be its image under the Hodge splitting. Let $\eta \in H^1(\Pi_f)_{\mathbb{C}}$ corresponding to a $K_\infty$-homomorphism $F_\eta : \tau_1 \to \Pi$. Then,*

$$\langle \iota_*^{\mathrm{nearly}}(E), \widetilde{\eta} \rangle = \frac{1}{(2\pi i)^2} \int_{Z_G(\mathbb{A})H(\mathbb{Q})\backslash H(\mathbb{A})} F_\eta(v_{\kappa_1,\kappa_2}) E_\infty(h) dh.$$

*Proof.* The proof of this requires that the Chevalley–Eilenberg complex for $(\mathfrak{p}, K)$-cohomology is concentrated in degree 1 and is represented by the minimal $K$-type, which is the content of [LPSZ, Lemma 6.5]. In our setting, this is the following statement:

$$\dim_{\mathbb{C}} \mathrm{Hom}_{K_\infty}\left(\wedge^j(\mathfrak{p}^-) \otimes \left(\widetilde{L}_1\right)^\vee, \Pi_\infty^{\mathrm{gen}}(k_1,k_2,k_3)\right) = \begin{cases} 0 & \text{if } j \neq 1, \\ 1 & \text{if } j = 1. \end{cases}$$

The proof of this can be done by comparing the $K$-types of the irreducible algebraic $G$-representation and that of the generic discrete series. With the aid of the following picture, we can prove the statement by noticing the following facts.




- In the above picture, $\alpha_2, -\alpha_2$ are the compact roots.
- The generic discrete series has $K$-types lying in the cone with a vertex on the minimal $K$-type and the rays to the directions of $\alpha_1+\alpha_2$ and $-\alpha_2$.
- The minimal $K$-type of the generic discrete series is $\alpha_1$ plus the rightmost $K$-type of $\widetilde{L}_1^\vee$. Thus, not only $\widetilde{L}_1^\vee$ has no intersection with generic discrete series, but also $\widetilde{L}_1^\vee \otimes \mathbb{C}\alpha_1$ contains the minimal $K$-type of generic discrete series with multiplicity one, and $\widetilde{L}_1^\vee \otimes \mathbb{C}(\alpha_1+\alpha_2)$ has no $K$-type lying in the region of generic discrete series.

□

6.4. **Automorphic and $p$-adic period pairing of nearly holomorphic modular forms.** We would like to develop the analogous theory for $p$-adic modular forms. To do this, we need a $p$-adic analogue of Hodge splitting over $\mathfrak{X}_G^{\geq 1}(p^m)$. The same proof as that of [LPSZ, Proposition 6.8] yields the following

**Proposition 6.11.** *Let $[\alpha] \in H^0(\mathfrak{IG}_G^{\geq 1}(p^\infty), \omega_A^\vee)$ be the natural morphism of $p$-divisible groups $\alpha : \mu_{p^\infty} \hookrightarrow A[p^\infty]$. Then, there is a unique lifting $[\alpha]^{\mathrm{nr}} : H^0(\mathfrak{IG}_G^{\geq 1}(p^\infty), \mathcal{H}^1_{\mathrm{dR}}(A)^{\mathrm{can}})$ such that its restriction to $\mathfrak{IG}_G^{\mathrm{ord}}(p^\infty)$ takes values in the unit root subsheaf $\mathcal{U} \subset \mathcal{H}^1_{\mathrm{dR}}(A)^{\mathrm{can}}$.*

**Definition 6.12.** Let $Q \subset P \cap \overline{B}$, where $\overline{B}$ is the lower-triangular Borel.

This is a parabolic subgroup of $M_P$, whose Levi is identified with $T$.

**Definition 6.13.** Let $\mathcal{T}^{\mathrm{Kl}}$ be the $Q$-torsor over $\mathfrak{X}_G^{\geq 1}(p^\infty)$ parametrizing bases $f_1, f_2, f_3$ of $\mathcal{H}^1_{\mathrm{dR}}(A)^{\mathrm{can}}$ compatible with the filtration and the polarization, with the additional property that the pullback of $f_1$ to $\mathfrak{IG}_G^{\geq 1}(p^\infty)$ is a scalar multiple of $[\alpha]^{\mathrm{nr}}$.

This is a reduction of structure of the $P$-torsor $\mathcal{T}$. Using this torsor, one can define the unit root splitting over $\mathfrak{X}_G^{\geq 1}(p^\infty)$ as follows.

**Proposition 6.14.**

*(1) For any $k_1 \geq -k_3$, there is a unique-up-to-scalars $Q$-equivariant surjection*

$$W_G(k_1, k_2, k_3; c) \to \lambda(k_1, k_2, k_3; c).$$

*(2) For any $k_1 \geq k_2 \geq -k_3$, there is a unique-up-to-scalars $Q$-equivariant surjection*

$$V_G(k_1, k_2, k_3; -k_1 - k_2 + k_3) \to \lambda(k_1 - k_2 - k_3, k_2, k_3; -k_1 - k_2 + k_3),$$

*which factors through $V/\operatorname{Fil}^{\frac{2k_1+k_2-3k_3}{2}+1} V$. The restriction of this morphism to $\operatorname{gr}^{\frac{2k_1+k_2-3k_3}{2}} V$ is nontrivial, and this factors through the map of form (i).*

*(3) For any $k_1 \geq k_2 \geq -k_3$, one has the "unit root splitting",*

$$\begin{tikzcd}
H^1(\mathfrak{X}_G^{\geq 1}(p), [\widetilde{L^\vee}_1](-D)) \arrow[dr, dashed] & \\
H^1(\mathfrak{X}_G^{\geq 1}(p), [L_1^\vee](-D)) \arrow[u, hook] \arrow[r] & H^1(\mathfrak{X}_G^{\geq 1}(p), \mathfrak{F}^{(k_1-k_2-k_3+1, k_2-1, k_3; k_3-k_1-k_2)}(-D)).
\end{tikzcd}$$

*(4) For $k_1 \geq \kappa_1 \geq k_2 \geq \kappa_2 \geq -k_3$, the following diagram commutes.*

$$\begin{tikzcd}
H^0(\mathfrak{X}_H^{\mathrm{ord}}(p^\infty), [V_H(\kappa_1, \kappa_2) \otimes \omega(1, -1)]) \arrow[r] \arrow[d] & H^1(\mathfrak{X}_G^{\geq 1}(p^\infty), [\widetilde{L_1^\vee}](-D)) \arrow[dr] & \\
H^0(\mathfrak{X}_H^{\mathrm{ord}}(p^\infty), \omega(\kappa_1 + 1, \kappa_2 - 1)(-D)) \arrow[r] & H^1(\mathfrak{X}_G^{\geq 1}(p^\infty), [L_1^\vee](-D)) \arrow[r] & H^1(\mathfrak{X}_G^{\geq 1}(p^\infty), \mathfrak{F}^{(k_1-k_2-k_3+1, k_2-1, k_3; k_3-k_1-k_2)}(-D)).
\end{tikzcd}$$

*Proof.* The proof is entirely analogous to that of Lemma 6.11, Corollary 6.15, and Theorem 6.16 of [LPSZ]. Namely, (i) is clear from the fact that $T$ is the Levi of $Q$, (ii) is because the target weight is an extremal weight, and (iii) is because of the construction of (ii). The part (iv) follows from the fact that the $Q$-torsor $\mathcal{T}^{\mathrm{Kl}}$ pulls back to a $Q$-torsor $\iota_\infty^*(\mathcal{T}^{\mathrm{Kl}})$ over $\mathfrak{X}_H^{\mathrm{ord}}(p^\infty)$, which has a reduction of structure to a $T$-torsor $\mathcal{T}^{\mathrm{ord}}$. □

From this, the compatibility of the unit root splitting and the Hodge splitting follows. Namely, Proposition 6.5 can be relaxed so that $\phi(\mathcal{E})$ can be the image of a classical *nearly holomorphic* modular form.

## 7. Local calculations: $\mathrm{GU}(2,1)$

We now conduct the calculations of the local zeta integrals for the Gelbart–Piatetski-Shapiro integral for $\mathrm{GU}(2,1)$ and the new zeta integral of GGP form for $\mathrm{GU}(2,1) \times \mathrm{GU}(1,1)$, insomuch that one can use them to conduct an algebraic and $p$-adic study of relevant $L$-functions. For the $\mathrm{GU}(2,1)$, the calculations are more or less all previously done in the literature.

7.1. **Nonarchimedean $L$-factors.** Let $F$ be a nonarchimedean local field over $\mathbb{Q}_v$, and let $E/F$ be a quadratic extension. Let $\pi_v$ be an irreducible admissible representation of $G(F)$ which is also *generic*. Recall the definition of the local zeta integral,

$$Z_v(f, \mu_v, \Phi_v, s) = \int_{Z_G(\mathbb{Q}_v) Z_{N_{B_G}}(\mathbb{Q}_v) \backslash H(\mathbb{Q}_v)} W_v(h) f_v^{\Phi}(h; \mu, s) dh,$$

where

$$f_v^{\Phi}(h; \mu, s) = \prod_{w|v} \int_{K_w^\times} (h \cdot \Phi_w)(ty) \mu_w(t) |t|_w^s d^\times t,$$

and $W_{f,v}(h)$ corresponds to $f$ in the local Whittaker model for $\pi_v$ with respect to the character $\psi_v$. Here the choice of an isomorphism between the local Whittaker model and $\pi_v$ is chosen and fixed implicitly. The following local calculation is performed in [GPS].

**Theorem 7.1.** *(1) There is a positive real number $R > 0$ depending on $\pi_v, \mu_v$ such that the local zeta integral $Z_v(f, \mu_v, \Phi_v, s)$ converges absolutely for $\mathrm{Re}(s) > R$ for all choices of $f, \Phi_v$. Furthermore, each $Z_v(f, \mu_v, \Phi_v, s)$ has a meromorphic continuation with respect to $s$, and is a rational function of $q_v^s$.*

*(2) The vector space of functions on $\mathbb{C}$ spanned by $Z_v(f, \mu_v, \Phi_v, s)$, as the data $(f, \Phi_v)$ vary, is a fractional ideal of $\mathbb{C}[q_v^{\pm s}]$ containing the constant functions. The fractional ideal is generated by $L(s, \pi_v, \mu_v)$, which is the local $L$-factor corresponding to the standard $L$-function for a $\mathrm{GL}_3(E)$-representation $BC(\pi_v) \otimes \mu_v$, the base-change lift of $\pi_v$ to $E$ tensored with $\mu_v$.*

*(3) If everything in the data is unramified, $f_v^0 \in \pi_v$ is the unique spherical vector such that $W_{f_v^0,v}(1) = 1$, and $\Phi_v^0 = \mathbf{1}_{\mathcal{O}_E^2}$, then*

$$Z_v(f_v^0, \mu_v, \Phi_v^0, s) = L(s, \pi_v, \mu_v).$$

*Proof.* This is precisely the content of [GPS, §4.4]. □

7.2. **Calculations at $p$.** We will be concerned with the "Klingen-ordinary" representations, which are in particular principal series representations as we saw in §3.3. The local zeta integral is that of $\mathrm{GL}_3 \times \mathrm{GL}_2$ by the hypothesis that $p$ splits in $K$. Let $\pi = I(\chi)$ be a spherical principal series of $\mathrm{GL}_3(\mathbb{Q}_p)$.

**Definition 7.2.** For $r \geq 1$, let $\phi_r \in \pi^{\mathrm{Kl}(p^r)}$ be the characteristic function with support $B(\mathbb{Q}_p)\,\mathrm{Kl}(p^r)$, scaled so that $\phi_r(1) = p^{2r}$.

A straightforward calculation gives the following

**Lemma 7.3.** *Let $r \geq 1$.*

*(1) The functions $\phi_r$ are compatible with taking traces, namely*

$$\frac{1}{[\mathrm{Kl}(p^r) : \mathrm{Kl}(p^{r+1})]} \sum_{\gamma \in \mathrm{Kl}(p^r)/\,\mathrm{Kl}(p^{r+1})} \gamma \phi_{r+1} = \phi_r.$$

*(2) The vector $\phi_r \in \pi^{\mathrm{Kl}(p^r)}$ is an eigenvector for $U_{p,1}$ with eigenvalue $p\chi(p)$.*

*(3) The vector $\phi_r$ satisfies*

$$\phi_r = \chi(p)^{1-r} \sum_{a \in \mathbb{Z}/p^{r-1}\mathbb{Z}} \begin{pmatrix} 1 & & p^r a \\ & 1 & \\ & & 1 \end{pmatrix} \begin{pmatrix} p & & \\ & p & \\ & & 1 \end{pmatrix}^{r-1} \phi_1.$$

**Theorem 7.4.** *Let $\Phi \in \mathcal{S}^2(\mathbb{Q}_p^2)$. Then, for $r \gg 0$, the zeta integral*

$$Z_v(\gamma \cdot \phi_r, \mu, \Phi, s) = \int_{Z_G(\mathbb{Q}_v) Z_{N_{B_G}}(\mathbb{Q}_v) \backslash H(\mathbb{Q}_v)} W_{\phi_r}(h\gamma) f^{\Phi}(h; \mu, s) dh, \quad \gamma = \begin{pmatrix} & & 1 \\ & 1 & \\ 1 & & \end{pmatrix},$$

*stabilizes to a value*

$$\frac{L(BC(\pi), s)}{L(BC(\pi^\vee), 1-s)\epsilon(BC(\pi), s)} \int_{\mathbb{Q}_p^\times} W_\xi(\left(\begin{smallmatrix} x & 0 \\ 0 & 1 \end{smallmatrix}\right)) W^{\Phi}(\left(\begin{smallmatrix} x & 0 \\ 0 & 1 \end{smallmatrix}\right), \mu, s) \frac{\theta(x)}{|x|} d^\times x.$$

*Proof.* The integral is exactly the Rankin–Selberg integral for $\mathrm{GL}_3 \times \mathrm{GL}_2$ whose local $p$-factor is calculated in [Ja, §4]. □

7.3. **Calculations at $\infty$.** This is one of the main purposes of [KO]. In *op. cit.*, one explicitly obtains the generalized Whittaker functions via solving differential equations arising from the Schmid operators. More precisely, we know that the Schwartz space is spanned by

$$\Phi_{(m_1,n_1,m_2,n_2)}(z_1, z_2) = z_1^{m_1} \overline{z}_1^{n_1} z_2^{m_2} \overline{z}_2^{n_2} \exp\left(-\frac{1}{2}\left(|z_1|^2 + |z_2|^2\right)\right), \quad m_1, n_1, m_2, n_2 \in \mathbb{Z}.$$

**Theorem 7.5** [KO, §6]**.** *Let $W$ be the Whittaker function of the generic discrete series $\Pi_\infty^{\mathrm{gen}}(\lambda_1, \lambda_2)$ contained in a $K$-type subspace of weight $(\lambda_1 + \lambda_2 + 2, -\lambda_1 - 1) + e(1,1) + f(1,-2)$. Then, $Z_\infty(W, \mu_\infty, \Phi_{(m_1,n_1,m_2,n_2)}, s) \neq 0$ if and only if $\omega_1 = n_1 + n_2 - m_1 - m_2$, $n_1 - m_1 = \lambda_1 + e$, $n_2 - m_2 = -e + f$, where $\omega_\infty(t) = |t|^{\omega_0} \left(\frac{t}{|t|}\right)^{\omega_1}$. Furthermore, it is equal to $C \cdot L(BC(\Pi_\infty^{\mathrm{gen}}(\lambda_1, \lambda_2)), s)$, where $C$ is a constant depending on $\lambda_1, \lambda_2$ only.*

## 8. Construction of the $p$-adic $L$-function

We now gather all the ingredients to construct the desired $p$-adic $L$-function.

8.1. **Family of Eisenstein series.** Our Eisenstein series lives over $\mathrm{U}(1,1)$ which is not so different from the case of elliptic modular forms. In fact, a holomorphic $\mathrm{U}(1,1)$-automorphic form is a holomorphic modular form, and the Shimura variety for $\mathrm{U}(1,1)$ is a finite union of modular curves. Therefore, the rationality property of $\mathrm{U}(1,1)$-Eisenstein series holds exactly as in [LPSZ, §7].

8.2. **Main theorem.** We retain the notations used in previous section.

**Theorem 8.1** (Theorem 1 of Introduction)**.** *Let $R = L[[\mathbb{Z}_p^\times]]$. If $k_1 - k_2 \geq 2$ and $k_2 + k_3 \geq 2$, there is an element $\mathcal{L}(\Pi, \mathbf{j}) \in R$ such that, for any Dirichlet character $\chi$ of $p$-power conductor and any critical value $a$ of $L(\Pi, \chi, s)$,*

$$\mathcal{L}(\Pi, a + \chi) = \frac{Z_p(\Pi, \chi, a) \Lambda(BC(\Pi) \otimes \chi^{-1}, a)}{\Omega^W(\Pi)},$$

*where*

$$Z_p = \frac{1}{L_p(BC(\pi^\vee), \chi^{-1}, 1-a)\epsilon(BC(\pi), a)},$$

*and $\Omega^W(\Pi)$ is the* Whittaker period*, defined in a way that the Whittaker function $W_f$ corresponding to an $E$-rational vector $f \in H^1(\Pi_f)$ is such that $\Omega^W(\Pi) W_f$ is an $E$-rational Whittaker function.*

*Proof.* We have shown that a choice of finite linear combination of $E$-rational Schwartz functions away from $\{v, \infty\}$ would give rise to the desired local $L$-factor. We have also shown how to choose the data at $v$ and $\infty$, respectively. Finally, there must be a factor of Whittaker period coming from the archimedean zeta integral, as it involves the value of Whittaker function at 1. □

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